

FKBP35 secures ribosome homeostasis in *Plasmodium falciparum*

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Abstract

Plasmodium falciparum accounts for the majority of over 600'000 malaria-associated deaths annually. Parasites resistant to nearly all antimalarials have emerged and the need for drugs with alternative modes of action is thus undoubted. The FK506-binding protein *Pf*FKBP35 has gained attention as a promising drug target due to its high affinity to the macrolide compound FK506 (tacrolimus). Whilst there is considerable interest in targeting *Pf*FKBP35 with small molecules, a genetic validation of this factor as a drug target is missing and its function in parasite biology remains elusive. Here, we show that limiting *Pf*FKBP35 levels are lethal to *P. falciparum* and result in a delayed-death phenotype that is characterized by defective ribosome homeostasis and stalled protein translation. We furthermore show that FK506, unlike the role of this drug in model organisms, exerts its anti-proliferative activity in a *Pf*FKBP35-independent manner and, using cellular thermal shift assays, we identify FK506-targets beyond *Pf*FKBP35. In addition to revealing first insights into the function of *Pf*FKBP35, our results show that FKBP-binding drugs can adopt non-canonical modes of action – with major implications for the development of FK506-derived molecules active against *Plasmodium* parasites and other eukaryotic pathogens.

eLife assessment

This **important** study addresses both the native role of the *Plasmodium falciparum* protein *Pf*FKBP35 and whether this protein is the target of FK506, an immunosuppressant with antiparasitic activity. The genetic evidence for the essentiality of FKBP35 in parasite growth is **compelling**. However, the conclusion that the role of FKBP35 is to secure ribosome homeostasis and the claim that FK506 exerts its antimalarial activity independently of FKBP35 rely on **incomplete** evidence.

1 Introduction

Despite considerable progress in recent years, malaria remains one of the major global health threats (1). The apicomplexan parasite *Plasmodium falciparum* is responsible for the vast majority of the over 600'000 malaria deaths in 2020. Most of this burden is carried by infants and children under the age of five in sub-Saharan Africa. Malaria deaths increased by 12% compared to 2019, which can at least in part be explained by the COVID-19 related suspension of malaria control and treatment measures (1). In addition, the emergence of parasites resistant to most antimalarials, including artemisinin-based combination therapies, endangers current and future disease elimination efforts and underscores the need for developing drugs with alternative modes of action (2).

FK506 (tacrolimus), a well-characterized immunosuppressant, shows considerable activity against asexual blood stage parasites (3). *P. falciparum* encodes a single FK506-binding protein (FKBP) dubbed *PfFKBP35* (PF3D7_1247400), which is considered to be a promising drug target (4; 5). FKBP's belong to the immunophilin family and are conserved throughout the eukaryotic kingdom (6; 7). In contrast to the malaria parasite, most other eukaryotes encode several FKBP's that, besides showing high affinity for the FK506 drug, exert peptidyl-prolyl isomerase (PPIase) activity. The PPIase moiety catalyzes cis-trans isomerization of proline residues – an event thought to be rate-limiting for the folding of many proteins (8). Independent of their PPIase activity, FKBP's act as chaperones to prevent protein aggregation under stress conditions (9; 10). Furthermore, some FKBP variants play a role in cellular signaling or gene regulation (11; 12). Human FKBP12 for instance is best-known for its ability to form a complex with the immunosuppressant drugs FK506 and rapamycin. These complexes inhibit the phosphatase activity of calcineurin and the kinase activity of mTOR (mechanistic target of rapamycin), respectively (13). Amongst many other effects, this inhibition results in decreased T-cell activation. Of note, the regulatory role of *HsFKBP12* depends on the presence of rapamycin or FK506. In absence of drugs, this immunophilin neither interacts with mTOR nor with calcineurin and does hence not affect activity of these key components of eukaryotic cells. Due to the conserved nature of FKBP's, rapamycin and FK506 have also gained attention as potential antimicrobial drugs (3; 7).

Similar to its homologs in model organisms, *PfFKBP35* harbors PPIase and chaperoning activity *in vitro* (5). Despite sharing key features with FKBP's of other eukaryotes, its role in *P. falciparum* remains elusive. *PfFKBP35* is believed to interact with heat shock proteins (HSPs) and co-immunoprecipitates with cytoskeletal factors of *P. falciparum* (14). While *PfFKBP35* is inhibiting the phosphatase activity of recombinant calcineurin in a FK506-independent manner ((15)–(17)), it does not co-localize with calcineurin *in vivo* (16). The protein is expressed throughout the 48-hour intra-erythrocytic development of *P. falciparum* and is found within the cytosol of ring stage parasites. In the older trophozoite and the replicating schizont stages, the protein was reported to also localize to the nucleus (16). While a transposon-based mutagenesis screen suggests that *PfFKBP35* is essential for intra-erythrocytic replication (18), neither its cellular function nor its essentiality was confirmed experimentally in live parasites. The unknown role in parasite biology notwithstanding, *PfFKBP35* is considered to be a viable drug target and, given the fact that *P. vivax* and *P. knowlesi* encode FKBP homologs that exhibit comparable PPIase activities (19; 20), future FKBP-targeted therapies may be effective across different *Plasmodium* species.

Crystal structures of *PfFKBP35* in complex with FK506 and rapamycin revealed high affinity interactions with the FK506-binding domain of the protein (15; 21; 22). Consistent with these observations, *in vitro* enzyme activity assays showed that FK506 inhibits the PPIase activity of recombinant *PfFKBP35* in a dose-dependent manner (5). Previous research aimed at exploiting structural differences between human and parasite-encoded FKBP's to circumvent

the immunosuppressive activity of drug-bound *HsFKBP12* ((23)–(26)). These efforts include investigation of non-covalent FKBP inhibitors such as adamantyl derivatives, macrocycles, the small molecule D44 ((23)–(25), (27), (28)) and the synthetic ligand for FKBP (SLF) as well as derivatives thereof designed to covalently bind *PfFKBP35* (29). While D44 shows promising antimalarial activity, it fails to alter thermostability of the FK506-binding domains of *PfFKBP35* and *HsFKBP12* *in vitro* (29), suggesting that this compound does not directly interact with *PfFKBP35*. Up until now, *PfFKBP35* has not been validated as a drug target using reverse genetics, and the link between *PfFKBP35*-interacting drugs and their antimalarial activity remains elusive (3).

Here, we generated inducible *PfFKBP35* knock-out, knock-down as well as over-expression cell lines and show that *PfFKBP35* is essential for asexual replication of blood forms. We demonstrate that *PfFKBP35* is vital for ribosome homeostasis on the post-transcriptional level and likely acts as a chaperone during the biogenesis of ribosomes in the nucleus. As a result, *PfFKBP35* knock-out parasites stall protein translation prior to dying at the early schizont stage, which underpins the potential of *PfFKBP35* as a drug target. Using cellular thermal shift assays (CETSA), we corroborate the high affinity binding of FK506 to *PfFKBP35* and reveal interactions of the *PfFKBP35*/FK506 pair with ribosome-associated proteins. Importantly, however, our data demonstrate that FK506 fails at inhibiting the essential function of *PfFKBP35* and exerts its antimalarial activity in a *PfFKBP35*-independent manner. While this has major implications for the development of future FK506-based antimalarials and/or *PfFKBP35*-inhibiting drugs, we identify putative targets of FK506 beyond *PfFKBP35* in *P. falciparum*.

2 Results

2.1 *PfFKBP35* is essential for asexual replication

In a first step, using CRISPR/Cas9-mediated gene editing, we engineered a *P. falciparum* cell line allowing for a conditional knock-down of *PfFKBP35* using the DD/Shield system (30; 31) (Figs. 1A and S1A–B). Despite the efficient depletion of *PfFKBP35* under knock-down conditions, parasites did not show apparent growth defects (Figs. S1C–D). Immunofluorescence assays (IFA) showed that *PfFKBP35* localizes to distinct foci within the parasite nucleus (Fig. 1B).

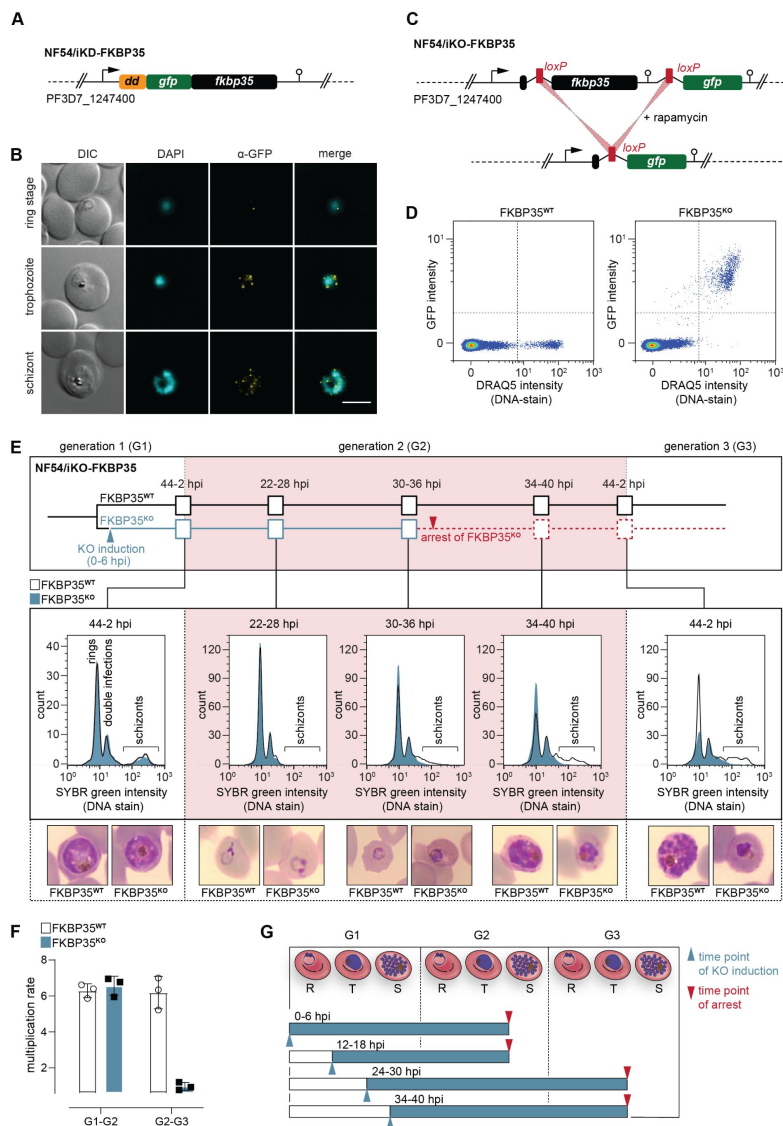


Figure 1.

*Pf*FKBP35 depletion causes delayed parasite death.

(A) Schematic of the N-terminally modified *fkbp35* locus in NF54/iKD-FKBP35 parasites. *dd*, destabilization domain. *gfp*, green fluorescent protein. (B) Localization of *Pf*FKBP35 in NF54/iKD-FKBP35 parasites at the ring, trophozoite and schizont stage assessed by immunofluorescence. DNA was stained with DAPI. Representative images are shown. Scale bar: 5 μ m. DIC, differential interference contrast. (C) Schematic of the modified *fkbp35* locus in NF54/iKO-FKBP35 parasites before and after *loxP* recombination. *gfp*, green fluorescent protein. (D) Flow cytometry plot showing GFP expression in G1 schizonts of FKBP35^{KO} and FKBP35^{WT}. DNA was stained with DRAQ5. (E) Time course showing the development of NF54/iKO-FKBP35 parasites. Upper panel: Schematic of the experimental setup. Middle panel: DNA content of the populations assessed by flow cytometry based on the SYBR Green intensity. Lower panel: Giemsa-stained blood smears. The knockout was induced at 0-6 hpi. Representative data of three biological replicates are shown. (F) Multiplication rates of FKBP35^{KO} and FKBP35^{WT} parasites from generation 1 to generation 2 (G1-G2) and G2-G3. The knockout was induced at 0-6 hpi. Multiplication rates were determined using flow cytometry.

Data points represent the multiplication rates from G1-G2 and G2-G3. $n=3$, error bars represent the standard error of the mean. (G) Schematic showing time point of growth arrest (red arrow) following *Pf*FKBP35 knock-out at different hpi (blue arrow). R, ring stage; T, trophozoite stage; S, schizont stage.

Next, we generated a cell line allowing for the conditional knock-out of the endogenous coding sequence using a dimerizable Cre recombinase (DiCre)-based system (32). Specifically, we introduced two *loxP* sites – one within a synthetic intron (33) downstream of the start codon and one after the stop codon of the *fkbp35* coding region – and added a green fluorescent protein (*gfp*) sequence downstream of this expression cassette (Figs. 1C, S2A). As a result, these NF54/iKO-FKBP35 parasites express tag-free, wild-type (WT) *Pf*FKBP35 protein under control conditions (FKBP35^{WT}). Upon DiCre activation, the *loxP* sites are recombined and the *fkbp35* locus is excised from the parasite genome (Figs. 1C, S2B). Under these knock-out conditions (FKBP35^{KO}), parasites express cytosolic GFP under control of the endogenous *fkbp35* promoter. This setup allows monitoring knock-out efficiency at the single cell level and revealed that DiCre successfully deleted *fkbp35* in the vast majority of parasites (Fig. 1D).

In a first attempt, we induced the *fkbp35* knock-out in synchronous ring stage NF54/iKO-FKBP35 parasites at 0-6 hours post erythrocyte invasion (hpi), i.e. prior to the expected peak transcription of *fkbp35* (34). These FKBP35^{KO} parasites showed a delayed death phenotype: parasites completed intra-erythrocytic development in the first generation (G1) without noticeable effects (Fig. 1E) and entered the next generation (G2) with a multiplication rate similar to that of the FKBP35^{WT} control population (Fig. 1F). In G2, however, FKBP35^{KO} parasites arrested at the late trophozoite/early schizont stage (approximately 30-36 hpi) with many cells failing to enter DNA replication in the S-phase (Fig. 1E). To investigate this phenomenon in more detail, we induced the *fkbp35* knock-out at three different time points during the IDC. Similar to DiCre activation at 0-6 hpi, knocking out *fkbp35* at 12-18 hpi resulted in parasite death at the trophozoite/schizont stage of the subsequent generation. However, when induced at 24-30 hpi or 34-40 hpi in G1, parasites did not only complete the current IDC but also successfully passed through G2 before they eventually arrested in G3 (Fig. 1G). Considering that low *Pf*FKBP35 levels are sufficient to maintain the essential function of the protein, as observed with the NF54/iKD-FKBP35 parasites (see Fig. S1C-D), it is conceivable that proteins expressed between 0 and 24hpi, i.e. prior to deleting *fkbp35* from the genome, allow parasites to enter G3.

The delayed death of FKBP35^{KO} is reminiscent of a phenotype observed in apicoplast-deficient parasites (35; 36). We therefore tested if *Pf*FKBP35 is involved in apicoplast biology, but did not find evidence for an association with this organelle (Fig. S2C). Furthermore, given the reported interaction of *Pf*FKBP35 with heat shock proteins (14; 16), we also tested the parasites response to heat stress, but could not detect altered heat-susceptibility between FKBP35^{KO} and FKBP35^{WT} cells (Fig. S2D).

2.2 *Pf*FKBP35 is crucial for ribosome homeostasis and protein translation

To further elucidate the function of *Pf*FKBP35, we induced the *fkbp35* knock-out in early ring stages (0-6 hpi) and compared the proteomes of FKBP35^{KO} and FKBP35^{WT} parasites at the schizont stage of G1 (36-42 hpi) and at the trophozoite stage of the following generation (G2, 24-30 hpi) using a quantitative proteomics approach.

2.2.1 The knock-out of *Pf*FKBP35 causes accumulation of pre-ribosome components in G1

Consistent with the efficient deletion of *fkbp35* on the genomic level, FKBP35^{KO} schizonts of G1 showed a 19-fold (18.9 +/-5.7 s.d.) reduction of FKBP35 and GFP levels were increased by 84-fold (83.7 +/-38.5 s.d.) when compared to the control population (Fig. 2A, Table S1). Besides *Pf*FKBP35 and GFP, 50 parasite proteins were significantly deregulated in FKBP35^{KO} schizonts and 34 of them found at elevated levels compared to the FKBP35^{WT} control conditions. A gene ontology analysis using PANTHER (37) indicated enrichment of the “pre-ribosome” (fold enrichment = 30.4, p-value = 1.46E-5, 4/34 proteins) and “nucleolus” (fold enrichment = 6.77, p-value = 8.85E-4, 5/34 proteins) terms among these factors. While the pre-ribosome is a large subunit precursor of mature ribosomes, the nucleolus represents the primary site for the biogenesis of ribosomal subunits (38). In contrast to factors being more abundant in knock-out parasites, none of the 16 factors with lower abundance in FKBP35^{KO} cells were associated with ribosome-related processes (Table S1). Despite being phenotypically silent in G1, the accumulation of pre-ribosome and nucleolar components hints at perturbed ribosome maturation in these cells. Specifically, it is conceivable that the observed buildup of ribosome intermediates results from reduced anabolic activity, i.e. the stalling of nucleolar pre-ribosome assembly.

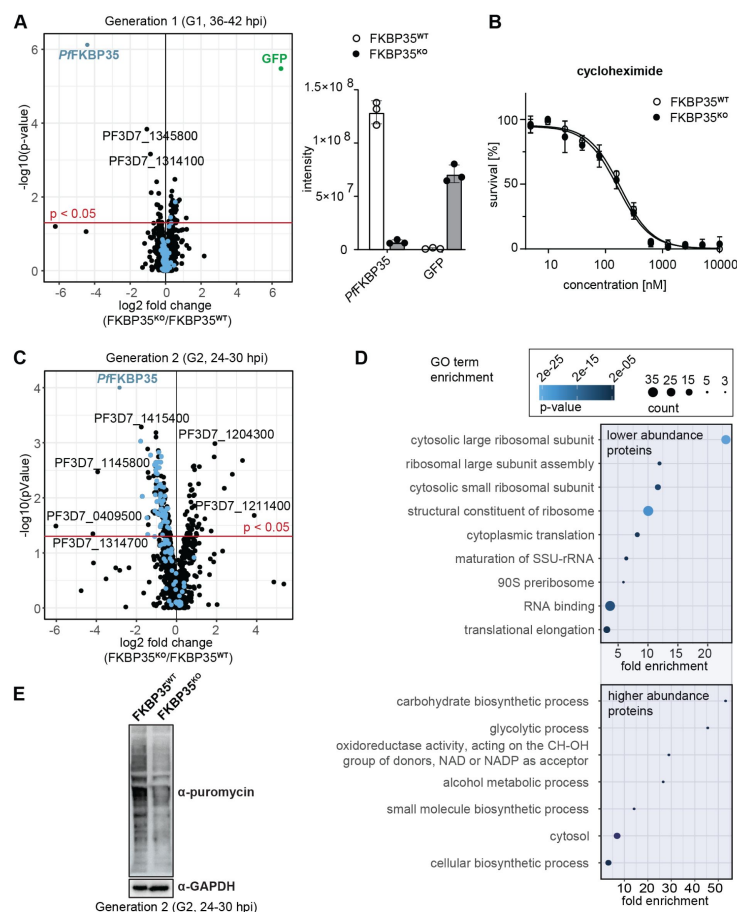


Figure 2.

***PfFKBP35* is crucial for ribosome homeostasis and protein translation.**

(A) Left panel: Proteome analysis of schizonts (36-42 hpi) in G1. The volcano plot shows change in protein abundance in *FKBP35*^{KO} compared to *FKBP35*^{WT}. Ribosomal proteins are highlighted in blue, n=3. Right panel: Relative difference in *PfFKBP35* and GFP abundance expressed as signal intensity measured by mass spectrometry. n=3, error bars represent the standard error of the mean. **(B)** Dose-response effect of the translation inhibitor cycloheximide on *FKBP35*^{KO} and *FKBP35*^{WT} parasites. The knockout was induced at 0-6 hpi and parasites were subsequently exposed to cycloheximide for 48 h. Parasite survival was quantified by flow cytometry. n=3, error bars represent the standard error of the mean. **(C)** Proteome analysis of trophozoites (24-30 hpi) in G2. The volcano plot shows change in protein abundance in *FKBP35*^{KO} compared to *FKBP35*^{WT}. Ribosomal proteins are highlighted in blue. n=3. **(D)** GO enrichment analysis of significantly deregulated proteins in *FKBP35*^{KO} trophozoites in G2. GO enrichment analysis was performed using the PANTHER “Overrepresentation Test” with the test type “Fisher’s Exact” and the correction method “False Discovery Rate”. **(E)** Assessment of translation activity using SunSET.

FKBP35^{KO} and *FKBP35*^{WT} trophozoites of G2 were incubated with puromycin for 1 hour. Incorporated puromycin was detected using an α-puromycin antibody. α-GAPDH served as loading control. n=3, a representative Western blot is shown.

Notwithstanding these effects, *FKBP35*^{KO} cells were equally sensitive to the translation inhibitor cycloheximide compared to their *FKBP35*^{WT} counterparts (Fig. 2B), indicating that ribosome function is not compromised at this point in time (39). This is consistent with normal cell cycle progression in G1 and unaltered multiplication rates of *FKBP35*^{KO} parasites, as described above (see Figs. 1E-F), and shows that the prominent depletion of *PfFKBP35* does not have a major impact on parasite viability in G1.

2.2.2 *PfFKBP35* is essential for maintaining normal levels of functional ribosomes in G2

In the following generation (G2, 24-30 hpi), the proteome of *FKBP35*^{KO} trophozoites showed substantial differences. Despite the absence of morphological changes at this point in time, 216 proteins were significantly deregulated (Fig. 2C). Among these, 50 proteins were found at higher abundance compared to *FKBP35*^{WT} and their gene ontology terms point towards an effect on processes involved in energy metabolism (Fig. 2D).

The majority of deregulated factors (166 of 216) in G2, however, were less abundant in *FKBP35*^{KO} compared to the *FKBP35*^{WT} control cells. Very prominently, 20 of the top 50 hit molecules are ribosomal proteins that, on average, show a two-fold reduction (2.0 ± 0.4

s.d.) in FKBP35^{KO} parasites (Table S1). It is therefore not surprising that PANTHER identified the knock-out of *Pf*FKBP35 to mainly affect ribosome-related processes, such as “cytoplasmic translation”, “ribosome assembly”, and “ribosomal large subunit biogenesis” (Fig. 2D) (37).

To assess if the reduced levels of ribosomal proteins result in lower protein translation rates, we employed the “surface sensing of translation” (SUnSET) technique. SUnSET is based on incorporation of the tyrosyl-tRNA analogue puromycin into nascent peptide chains and thereby allows monitoring newly synthesized proteins (40). As expected, we found that FKBP35^{KO} trophozoites in G2 (24-30 hpi) showed reduced translation rates compared to the FKBP35^{WT} control population (Fig. 2E). Further, protein synthesis was not affected in parasites of the parental line NF54/DiCre following DiCre activation using rapamycin (S2E), demonstrating that translation is impaired in response to knocking out *P*/FKBP35.

Taken together, these results indicate that, under knock-out conditions, *Pf*FKBP35 levels become limiting for ribosome maturation late during schizogony of G1, without having immediate effects on the steady-state levels of mature ribosomes. The knock-out of *Pf*FKBP35 is thus phenotypically silent in G1. In absence of a continuous supply of mature ribosomes under FKBP35^{KO} conditions, however, FKBP35^{KO} parasites fail at synthesizing protein at levels that would support normal parasite development in the next generation (G2).

2.3 The knock-out of *Pf*FKBP35 is transcriptionally silent

To examine if the altered proteome of FKBP35^{KO} parasites is the result of *Pf*FKBP35-controlled transcription, we compared the transcriptional profiles of FKBP35^{WT} and FKBP35^{KO} in a time course RNA-Sequencing (RNA-Seq) experiment spanning two IDCs. Specifically, we induced the knock-out of FKBP35 in synchronous parasites at 0-6 hpi and collected RNA from parasite populations at three and four subsequent time points (TPs) in G1 and G2, respectively (Figs. 3A, S3A). With the exception of TP7, these time points cover a time frame during which FKBP35^{KO} cells cannot be distinguished from their FKBP35-expressing counterparts on a morphological level (compare Fig. 1E).

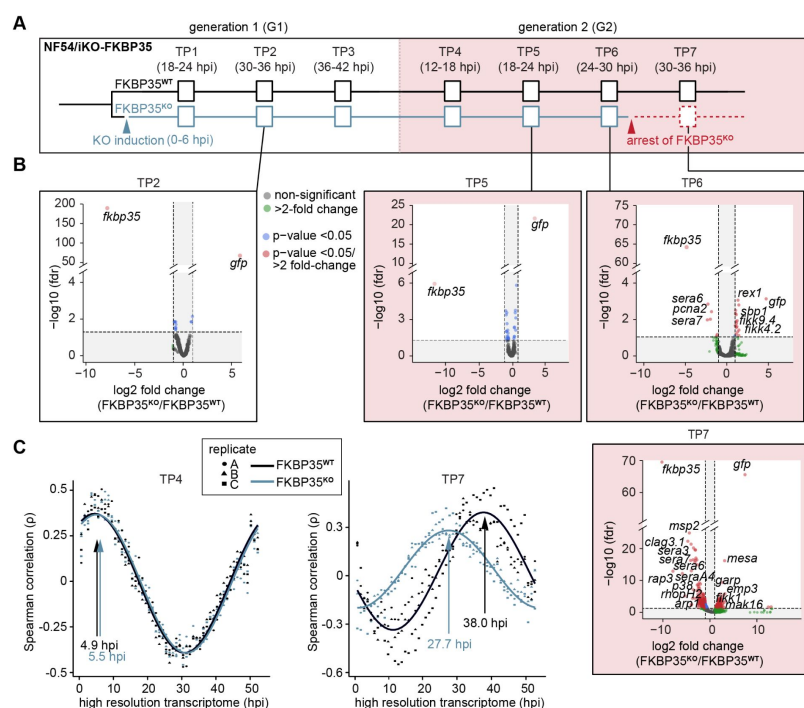


Figure 3.

The knockout of *PfFKBP35* remains silent at the transcriptomic level.

(A) Setup of the RNA-Seq experiment. RNA was collected at seven indicated time points (TP). **(B)** Volcano plots show differential gene expression between FKBP35^{KO} and FKBP35^{WT} parasites. Genes are color-coded according to the fold change and *fdr* (false discovery rate), *n*=3. **(C)** Mapping of FKBP35^{KO} and FKBP35^{WT} transcriptomes to a high-resolution reference (41). Spearman rank coefficients (*ρ*) show correlation between the sampled transcriptomes and the reference data set (41). The time points with the highest correlation coefficient are indicated.

As expected, the transcriptome of FKBP35^{KO} showed a prominent reduction of steady-state *fkbp35* transcripts already in TP1 (11.3-fold $\pm$ 2.0 s.d.) compared to the FKBP35^{WT} control population (Fig. S3B, Table S2). During peak transcription at 30-36 hpi (TP2), this difference increased to 240-fold (238.4 $\pm$ 20.0 s.d.) and, consistent with the efficient recombination at the *fkbp35* locus, these cells started transcribing *gfp*. Notwithstanding the prominent depletion of *fkbp35* transcripts, the transcriptome of FKBP35^{KO} is only marginally affected up to and including 24-30 hpi in G2 (TP6). By contrast, at 30-36 hpi (TP7) the transcription of knock-out parasites shows major changes (Fig. 3B). Considering the appearance of morphological changes in FKBP35^{KO} at this time point, effects on the transcriptome are unlikely to represent specific activities of *Pf*FKBP35 in gene regulation. Indeed, mapping the transcriptome of individual time points to a high resolution reference (41), revealed a significant slow-down of FKBP35^{KO} compared to FKBP35^{WT} parasites at TP7 (Figs. 3C, S3C). This indicates aborted cell cycle progression in FKBP35^{KO} parasites during a phase that is characterized by a rapid increase in biomass – possibly due to a translation and/or size-dependent cell cycle checkpoint (42). At earlier time points (TP1-TP6), the transcriptomes of FKBP35^{KO} and FKBP35^{WT} are comparable and the differences observed on the level of individual genes clearly fail at explaining the proteomic changes observed in FKBP35^{KO} parasites at 24-30 hpi in G2 (TP6), reiterating that the essential activity of *Pf*FKBP35 is linked to post-transcriptional processes.

2.4 FKBP-targeting drugs fail at inhibiting the essential function of *Pf*FKBP35

*Pf*FKBP35 emerged as a promising antimalarial drug target (4) and considerable efforts were made to define interactions of this immunophilin with small molecules in the last decade ((24)–(26), (43)). Several studies conclusively demonstrated that FK506 – the most well-known ligand of mammalian FKBP (44) – binds to *Pf*FKBP35 (15; 21; 22) and kills asexual blood stage parasites with a half-maximal inhibitory concentration (IC₅₀) in the low micromolar range at 1.9 μ M (3).

Considering the proposed direct interaction between FK506 and FKBP35 in *P. falciparum*, we expected that FKBP35^{KO} parasites show altered sensitivity to the drug. Consistent with published data, FK506 killed FKBP35^{WT} cells with an IC₅₀ of 1942 nM ($\pm$ 193 nM s.d.) when administered at 0-6 hpi (Fig. 4A). Surprisingly however, and despite the prominent depletion of the protein under knock-out conditions, FKBP35^{KO} and FKBP35^{WT} parasites were equally sensitive to FK506 (IC₅₀ = 1741 $\pm$ 187) (Fig. 4A). Similarly, neither the efficient knock-down of *Pf*FKBP35 in NF54/iKD-FKBP35 parasites (compare Fig. S1C), nor its over-expression (OE) in NF54/iOE-FKBP35 cells (Fig. S4A-C), resulted in the expected sensitivity change to FK506 treatment (Fig. 4B). In addition to this apparent independence of the drug to varying *Pf*FKBP35 levels, and in agreement with earlier studies (3; 24; 45), FK506 unfolds its antimalarial activity fairly quickly, i.e. within the same cycle after treatment. Standing in stark contrast to the delayed death phenotype of FKBP35^{KO} parasites (compare Fig. 1E-G), this immediate activity on parasite survival substantiates the notion that FK506 must target essential parasite factors other than *Pf*FKBP35.

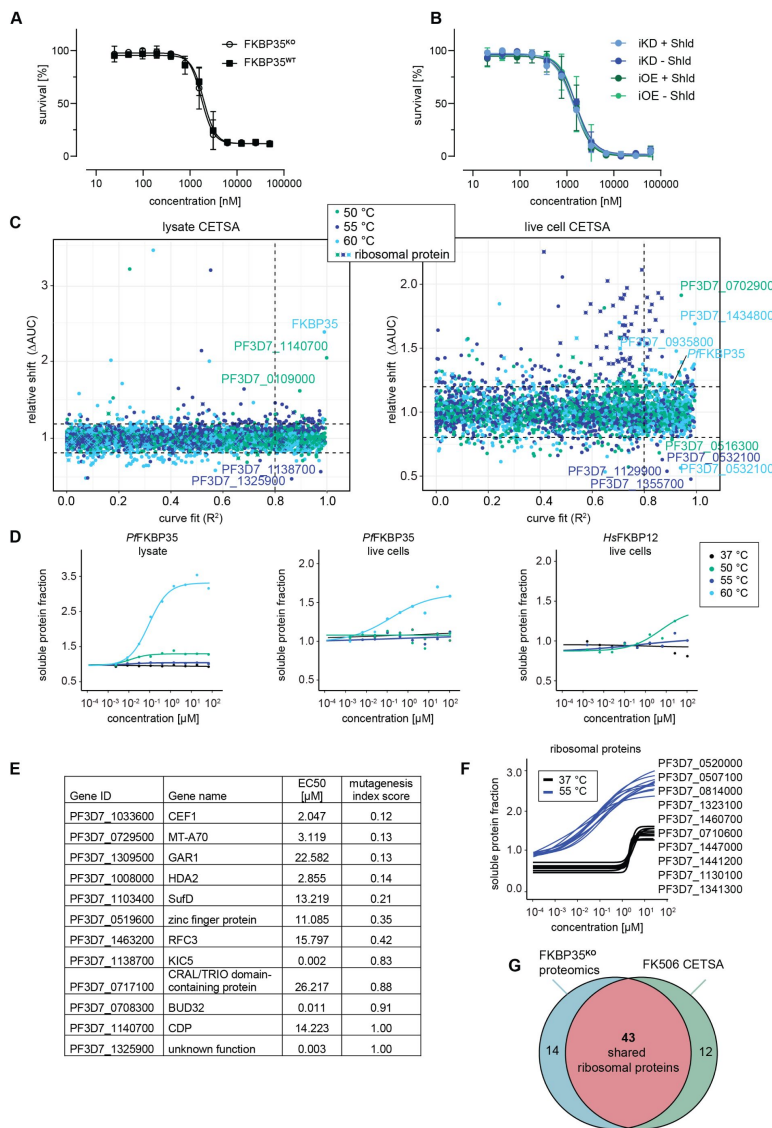


Figure 4.

CETSA identifies FK506 interaction partners.

(A) Dose-response effect of FK506 on FKBP35^{KO} and FKBP35^{WT} parasites. The knockout was induced at 0-6 hpi and parasites were exposed to FK506 for 48h. n=3, error bars represent the standard error of the mean. **(B)** Dose-response effect of FK506 on NF54/iKD-FKBP35 and NF54/iOE-FKBP35 parasites in presence and absence of Shield-1 (Shld). Parasites were split at 0-6 hpi and subsequently treated with Shld or the vehicle control EtOH. Data points represent the mean of three and two independent biological replicates for the NF54/iOE-FKBP35 and NF54/iKD-FKBP35 cell line, respectively. Error bars represent the standard error of the mean. **(C)** CETSA results. The relative shift in protein abundance (area under the curve of heat-challenged samples normalized against the non-denaturing control, ΔAUC) as a function of R² (goodness of fit as a measure for the dose-response effect). The dashed lines indicate an arbitrary cutoff of 3 times the median absolute deviation (MAD) of each dataset and R²=0.8. Three different temperatures were used for protein denaturation (indicated by colors). **(D)** Dose-dependent stabilization of *P. falciparum* and human FKBP35 detected using lysate and live cell CETSA. Bullets represent the abundance of non-denatured proteins in response to increasing FK506

concentrations under thermal challenge relative to the DMSO control. Colors indicate the temperatures used for heat challenge; the black curve represents the non-denaturing (37 °C) control. **(E)** Proteins with the most pronounced stability shift under thermal challenge in the lysate CETSA experiment are listed. The EC50 is the concentration at which the half-maximal effect of the FK506-induced thermal stabilization is reached. The mutagenesis index score determined by Zhang *et al.* (18) is a measure to predict if a protein is essential for asexual growth: 0=essential, 1=non-essential. **(F)** Fitted live cell CETSA curves of 10 representative ribosomal proteins at 37 °C and 55 °C. Ribosomal proteins show a FK506 dose-dependent increase in soluble protein fraction. **(G)** Venn diagram showing the overlap between ribosomal proteins deregulated in the *PfFKBP35* knock-out cell line (less abundant in G2 FKBP^{KO} trophozoites determined by proteomics) and the ribosomal proteins stabilized by FK506 in live cell CETSA. The number of factors identified in either or both assays is indicated.

To rule out that the timing of drug exposure is masking expected links between FK506 and *PfFKBP35*, we tested whether administering FK506 at 30-36 hpi (instead of 0-6 hpi) affects parasite survival in a *PfFKBP35*-dependent manner. However, we did not observe a change compared to FKBP35^{WT} parasites (Fig. S5A), ruling out the possibility that FKBP35^{KO} parasites were killed by FK506 before *PfFKBP35* levels were different in the two populations.

Further, to account for a scenario in which the activity of FK506 on *Pf*FKBP35 is masked by a temporal delay between drug exposure and parasite killing, we induced the knock-out of *Pf*FKBP35 at 34–40 hpi (instead of 0–6 hpi) and allowed FKBP35^{KO} parasites to proceed to G2, before exposing them to different FK506 concentrations. Despite these efforts, FKBP35^{WT} and FKBP35^{KO} parasites were equally sensitive to FK506 (Fig. S5B, S5C) and the calcineurin inhibitor cyclosporin A (Fig. S5D–E).

In addition to FK506, we evaluated the *Pf*FKBP35-specific inhibitor D44 (N-(2-Ethylphenyl)-2-(3h-Imidazo[4,5-B]pyridin-2-Ylsulfanyl)acetamide) but, surprisingly, we failed to reproduce the previously reported dose-response effect (24). We tested D44 from two suppliers using NF54/iKO-FKBP35, NF54 wild type and K1 parasite strains using different assay setups, including the gold standard method that is based on the incorporation of tritium labeled hypoxanthine into DNA of replicating cells (46). Still, we could not detect any activity of D44 on asexual replication of blood stage parasites (Figs. S5F–G).

Taken together, these results are in clear conflict with the perception that FK506 and D44 target the essential activity of *Pf*FKBP35.

2.5 Probing FK506 interactions in Cellular Thermal Shift Assays (CETSA)

Considering the unaltered activity of FK506 in FKBP35^{KO} compared to FKBP35^{WT} parasites, we reasoned that FK506 must target proteins other than *Pf*FKBP35. To identify potential targets of the drug in *P. falciparum*, we used the *cellular thermal shift assay followed by mass spectrometry* (MS-CETSA). This approach exploits the fact that, once drug-protein complexes are formed, thermostability of the target protein is altered (47). We applied the so-called *isothermal dose-response* variant of CETSA using FK506 (48). Here, target proteins are identified based on their dose-dependent stabilization by drugs under thermal challenge (different temperatures) when compared to a non-denaturing condition (37 °C), using the change in the area under the curve (Δ AUC) and the dose-response curve goodness of fit (R^2) as metrics (Fig. 4C) (48).

2.5.1 CETSA identifies targets of FK506

In a first step, we performed CETSA in combination with drug-exposed protein extracts. Since proteins lose their physiological context during the process of protein extraction, this variant of the CETSA approach is designed to identify direct protein-drug interactions (47; 48). Not surprisingly, we found *Pf*FKBP35 to be stabilized by FK506 in a dose-dependent manner when challenged at 60 °C with an EC₅₀ of 152.0 nM (Fig. 4D). The fact that FK506 exhibits its stabilizing activity already in the low nanomolar concentration range indicates high affinity interactions between the protein-drug pair. This FK506-mediated stabilization of *Pf*FKBP35 is thus achieved at drug concentrations that are magnitudes below the IC₅₀ of FK506 in parasite survival assays (1942 nM $\pm$ 193 nM s.d.) (compare Fig. 4A). In line with several previous reports on the *Pf*FKBP35/FK506 pair *in vitro*, our results show that, indeed, FK506 is binding to *Pf*FKBP35, but in contrast to previous assumptions this interaction fails at explaining the antimalarial activity of FK506.

Besides *Pf*FKBP35, FK506 stabilizes a number of other proteins in a dose-dependent manner (Figs. 4E and S6). A few of these factors show pronounced sigmoidal stabilization profiles and are predicted to be essential (Fig. 4E) (18). Amongst others, N6-methyl transferase MT-A70 (PF3D7_0729500), histone deacetylase HDA2 (PF3D7_1008000), pre-mRNA-splicing factor CEF1 (PF3D7_1033600), FeS cluster assembly protein SufD (PF3D7_1103400) and the H/ACA ribonucleoprotein complex subunit 1 (PF3D7_1309500) showed most pronounced stabilization profiles (Fig. S6 and Table S3). In contrast to *Pf*FKBP35, most of these likely

essential candidates are stabilized at FK506 concentrations close to the drug's IC₅₀ (compare Fig. 4A). It is therefore tempting to speculate that the binding of FK506 to at least one of these factors is responsible for the antimalarial activity of the drug.

2.5.2 The *Pf*FKBP35/FK506 pair interferes with ribosomal complexes

Given the high-affinity interaction of *Pf*FKBP35 with FK506 and its association with ribosomal proteins, we set out to describe the native context of *Pf*FKBP35 using the live cell variant of CETSA. In contrast to focusing exclusively on direct protein-drug interactions, live cell CETSA allows capturing drug-induced shifts in protein stability beyond those of direct binding partners (48) and amongst others allows identifying factors present in the same protein complex as the target protein. To do so, CETSA takes advantage of the fact that factors within complexes have similar denaturation temperatures due to their co-aggregation upon heat-induced denaturation (49). As a consequence, this CETSA variant generally identifies more factors compared to the protein extract-based CETSA approach (Fig. 4C).

As expected, and confirming the results above, live cell CETSA shows that FK506 alters the thermal stability of *Pf*FKBP35 already at very low drug concentrations when challenged at 60 °C with an EC₅₀ of 286.7 nM (Fig. 4D). Interestingly, and in contrast to the protein lysate-based approach, live cell CETSA revealed that more than 50 ribosomal proteins are stabilized in presence of FK506 when challenged at 55°C, with a median EC₅₀ of 45.9 nM (25th percentile = 34.2, 75th percentile = 58.9). Of note, these ribosomal proteins were stabilized at virtually identical FK506 concentrations (Figs. 4F, and S7), indicating that the drug – directly or indirectly – interacts with ribosomal complexes. This is in further agreement with the fact that these ribosomal factors were not identified by protein lysate-based CETSA (compare Figs. 4C, S6).

Worth mentioning, we observed a drug dose-dependent increase in soluble protein abundance of ribosomal factors also under non-denaturing conditions, i.e. at 37 °C, with a median EC₅₀ of 1.94 μM (25th percentile = 1.85, 75th percentile = 2.1) (Figs. 4F, S7). Generally, due to the absence of a denaturing heat challenge, such a drug-dependent enrichment in soluble protein levels does not indicate binding-induced thermal stabilization of proteins (48). Instead, this can be the result of increased protein synthesis, altered protein conformation or, alternatively, occur in response to factors disengaging from larger protein complexes. The fact that FK506 solubilizes a high number of ribosomal proteins within a very narrow concentration window also at 37 °C, implies that these factors are liberated from the same ribosomal complex or from complexes that interact with each other and shows that FK506 negatively affects stability of ribosomal structures in live cells. However, these effects are limited to high, presumably saturating FK506 concentrations, likely explaining why the drug fails at killing parasites in a *Pf*FKBP35-dependent manner. Importantly, the vast majority of ribosomal proteins identified in live cell CETSA are shared with those de-regulated in *Pf*FKBP35 knock out cells (Fig. 4G), providing conclusive evidence for a shared localization of FK506 and its *Pf*FKBP35 target in live parasites.

Together with the above-described importance of *Pf*FKBP35 in ribosome homeostasis, these results strongly suggest that FK506 is recruited to the activity site of *Pf*FKBP35 – most probably the nucleolar maturation sites of ribosomes – where it leads to a destabilization of ribosomal complexes.

3 Discussion

Members of the FKBP family are involved in regulating diverse cellular processes in eukaryotic as well as in prokaryotic cells (8; 50). In contrast to most other eukaryotes, the human malaria parasite *P. falciparum* encodes a single FKBP only (16). The enzymatic activity of *Pf*FKBP35, as well as the interactions of the protein with FKBP-targeting drugs, was subject to considerable drug-focused research *in vitro*. Besides demonstrating the drug's affinity to *Pf*FKBP35 by co-crystallization, these efforts uncovered that FK506 inhibits enzymatic activity of the protein's PPIase domain (5). Together with the anti-malarial activity of FK506, these data offered compelling evidence for an essential but yet uncharacterized role of *Pf*FKBP35 during blood stage development of *P. falciparum*.

Here, we set out to investigate the role of *Pf*FKBP35 in parasite biology. Using an inducible knock-out approach, we show that *Pf*FKBP35 is indeed essential for intra-erythrocytic development of asexually replicating *P. falciparum* parasites and FKBP35^{KO} cells die during the late trophozoite/early schizont stage. Surprisingly, these parasites display a delayed-death phenotype that is independent of apicoplast function and only manifests in the cycle following knock-out induction without affecting cell cycle progression in the first generation (G1) or multiplication rates from G1 to G2. This contradicts the known effects of FK506 and other *Pf*FKBP35-targeting drugs for two main reasons. First, these drugs exert their parasitocidal activity within the same IDC (24). Second, neither the conditional knock-out, knock-down nor the overexpression of *Pf*FKBP35 altered the parasite's sensitivity to FK506, even though levels of the *Pf*FKBP35 target are profoundly deregulated in the respective mutant lines. Together, these data show that the parasite-killing activity of FK506 is independent of the essential function of *Pf*FKBP35. We hence aimed at the identification of other FK506 targets using CETSA, as discussed further down.

Knocking out *Pf*FKBP35 early during intra-erythrocytic development has no effect on parasite morphology in G1 and provokes only moderate changes on the protein level compared to FKBP35^{WT} parasites. While the effects in G1 are relatively subtle, it is noticeable that nucleolus-associated factors are found at higher abundance in knock-out cells. In the next cycle, FKBP35^{KO} parasites show a dramatic decrease of factors associated with ribosome- and translation-related processes during trophozoite development (G2, 24-30 hpi). Consistently, protein translation in FKBP35^{KO} parasites is reduced considerably. Importantly, at this point in time, FKBP35^{KO} cells still show ordinary morphology and their transcriptional activity remains largely unaffected. The observed changes in the abundance of ribosomal factors must therefore result from post-transcriptional activity of *Pf*FKBP35. While ribosomal components are present at higher levels in G1, they are drastically reduced in FKBP35^{KO} cells in G2. These opposing effects are peculiar, but likely result from *Pf*FKBP35 activity in a shared cellular process. Specifically, it is conceivable that low *Pf*FKBP35 levels inhibit ribosome maturation in G1, explaining the accumulation of (pre-) ribosomal factors. While the levels of functional ribosomes are sufficient to support continued translation in G1, ribosomes likely become limiting in the next generation (G2), which eventually results in stalled translation activity and concomitant parasite death at the onset of schizogony; i.e. at a time when protein synthesis is required for facilitating the steep increase in biomass.

In contrast to the significant effects on the protein level, the transcription of FKBP35^{KO} remains largely unaffected by the loss of *Pf*FKBP35 and becomes apparent only during late trophozoite/early schizont development in G2 (30-36 hpi); i.e. at a time during which cell cycle progression is slowed down in FKBP35^{KO} parasites. In fact, at this stage the transcriptional signature of FKBP35^{KO} parasites is dominated by the slowed progression through intra-erythrocytic development, rather than resulting from specific gene regulatory events. While this stalling is consistent with the observed effects on ribosome homeostasis

and protein translation, it demonstrates that, in contrast to certain FKBP family members in other organisms, *Pf*FKBP35 does not have apparent roles in transcriptional control.

As discussed above, binding of FK506 to *Pf*FKBP35 cannot explain the drug's antimalarial activity and our attempts to identify targets of FK506 using resistance selection failed. This indicates that parasites do not readily acquire resistance mutations, possibly due to drug action on multiple targets (51). We hence aimed at identifying other FK506 targets using CETSA. In line with previous work on the drug *in vitro*, CETSA on *P. falciparum* protein extracts confirmed that FK506 is targeting *Pf*FKBP35 with exceptionally high affinity in the low nanomolar range. Considering this well-characterized FK506-*Pf*FKBP35 interaction (22) and the resulting inhibition of its PPIase activity (5), unaltered drug sensitivity of *Pf*FKBP35 mutants indicates that PPIase activity is dispensable for blood stage parasites. Noteworthy, PPIase-independent functions of FKBP have been described in other systems. For instance, PPIase activity of the *Escherichia coli* trigger factor (TF), a ribosome-associated FKBP, is not required for protein folding (52). Similarly, human FKBP52 maintains its function as a chaperone under PPIase-inhibiting conditions (9). While it is conceivable that PPIase activity is dispensable for intra-erythrocytic development of *P. falciparum*, the enzymatic activity of *Pf*FKBP35 may be compensated by other factors, e.g. by PPIase domain-harboring members of the cyclophilin family (53).

CETSA performed on protein lysates identified a number of parasite factors binding to FK506. Compared to the *Pf*FKBP35/FK506 pair, however, these interactions are of considerably lower affinity. The adenosine-methyltransferase MT-A70 (PF3D7_0729500) and the pre-mRNA-splicing factor CEF1 (PF3D7_1033600), for instance, were stabilized by FK506 with EC₅₀ concentrations of 2.0 and 3.1 μ M, respectively. While this affinity is relatively poor compared to the strong interaction formed between *Pf*FKBP35 and FK506, one should be cautious designating the binding of FK506 to these proteins as irrelevant “off target” effects. In fact, their inhibition may well be masked by high-affinity binding of FK506 to *Pf*FKBP35 – a possibility that shall briefly be discussed in the following section.

Assuming that FK506 is inhibiting the essential function of *Pf*FKBP35, one would expect parasites to show a delayed-death phenotype similar to that observed in FKBP35^{KO} parasites. In contrast to this delayed killing, the effect of FK506 on other essential targets likely manifests in the same cycle and would hence mask the effect of *Pf*FKBP35-inhibition. Consistent with this possibility, FK506 kills *P. falciparum* parasites at 1.9 μ M (+/-0.2 μ M s.d.), i.e. at concentrations similar to those required for stabilizing MT-A70 and CEF1 under thermal challenge in CETSA. Our attempts to prove that FK506 indeed inhibits essential functions of *Pf*FKBP35 failed: Neither knocking out *Pf*FKBP35 at different time points nor adding FK506 to different stages of FKBP35^{KO} parasites could provide evidence for a scenario in which *Pf*FKBP35-dependent effects are simply masked by “off target” effects (compare Fig. S5). This indicates that either the binding of FK506 does not interfere with the essential role of *Pf*FKBP35, or that *Pf*FKBP35 is inhibited only at high FK506 concentrations that also inhibit other essential factors. In this context it is worth mentioning that parasites remain unaffected by a highly efficient knock-down of endogenous *Pf*FKBP35 (observed with NF54/iKD-FKBP35; compare Figs. S1C-D), which demonstrates that very low protein levels are sufficient to maintain the essential function of *Pf*FKBP35. It is thus conceivable that high FK506 concentrations are required to fully inhibit *Pf*FKBP35 – a view that is further supported by data obtained with the live cell CETSA approach.

Besides identifying direct binding of drugs to their target, live cell CETSA allows capturing indirect effects on factors that act in the same pathway or in the same protein complex as the drug-bound target (48). While this approach confirmed the high affinity binding of FK506 to *Pf*FKBP35, it also revealed that more than 50 ribosomal proteins share a very prominent CETSA signature: First, they are thermally stabilized by FK506 within a narrow concentration window of the drug. Second, higher levels of FK506 increase the solubility of

these ribosomal components also at 37 °C. These results indicate that the FK506/*Pf*FKBP35 pair interacts with ribosomal complexes and that FK506 is able to induce the dissociation of these ribosomal structures under physiological conditions. Importantly, and in contrast to the binding of FK506 to *Pf*FKBP35, this effect on ribosomes is only observed at high concentrations of the drug. Together, these data likely explain the discrepancy observed between the high binding affinity of FK506 for *Pf*FKBP35 and the low or even absent effect of the drug on *Pf*FKBP35 function in blood stage parasites.

In line with our data, previous biochemical approaches revealed potential links between *Pf*FKBP35 and ribosomes. Using co-immunoprecipitation, Leneghan and colleagues revealed that *Pf*FKBP35 is interacting with 480 proteins, of which 31 are ribosomal factors (14). Indeed, FKBP family members are essentially involved in controlling ribosomes in other systems. The human FKBP variant *Hs*FKBP25, for instance, is recruited to the pre-ribosome where its chaperone activity appears to be required for the assembly of the ribosomal large subunit (54). While the FKBP variant of yeast, *Fpr4*, interacts with the ribosome biogenesis factor *Nop53* (55), the *Escherichia coli* trigger factor (TF) is part of the ribosomal complex and crucial for the folding of nascent proteins (56; 57). Considering the marked reduction of ribosomal factors in FKBP35^{KO} parasites, it is tempting to speculate that *Pf*FKBP35 is chaperoning ribosome biogenesis similar to *Fpr4* and FKBP25 in *Saccharomyces cerevisiae* and *humans*, respectively.

The fact that *Pf*FKBP35 localizes to distinct nuclear foci and its depletion causes dysregulation of ribosomes points towards a role within the nucleolus – the nuclear compartment driving ribosome biogenesis (38). Using IFAs, we found that *Pf*FKBP35 does not co-localize with the nucleolar marker fibrillarin (58) (Fig. S8). However, its localization appears to be intertwined with that of this rRNA methyltransferase. Specifically, *Pf*FKBP35 is mostly found in close vicinity to fibrillarin-rich regions. Nucleoli in eukaryotic cells generally consist of two or, in higher eukaryotes, three sub-compartments. Of those, the granular component (GC) represents the site of pre-ribosome assembly and encompasses fibrillarin-enriched central regions of nucleoli (59). The spatial correlation observed between *Pf*FKBP35 and fibrillarin may thus indicate localization to different sub-nucleolar compartments, with *Pf*FKBP35 occupying the GC area of the nucleolus.

In summary, we demonstrate that limiting *Pf*FKBP35 levels are lethal to *P. falciparum* and result in a delayed-death phenotype that is characterized by perturbed ribosome homeostasis and defective protein translation, which is likely linked to the reduced biogenesis of functional ribosomes in the nucleolus. We further show that FKBP-binding drugs, including FK506, exert their parasitocidal activity in a *Pf*FKBP35-independent manner, urging strong caution for the future development of FKBP-targeting antimalarials, especially when based on FK506 and structural derivatives thereof. In addition to revealing first insights into the essential function of *Pf*FKBP35 in ribosome homeostasis of *P. falciparum*, we are convinced that the presented data offer valuable information for target-based efforts in malaria drug discovery.

4 Methods

4.1 Parasite culture

P. falciparum culture and synchronization was performed as described (60; 61). Parasites were cultured in AB+ or B+ human red blood cells (Blood Donation Center, Zürich, Switzerland) at a hematocrit of 5% in culture medium containing 10.44 g/L RPMI-1640, 25 mM HEPES, 100 µM hypoxanthine, 24 mM sodium bicarbonate, 0.5% AlbuMAX II (Gibco

#11021-037), 0.1 g/L neomycin, and 2 mM choline chloride (Sigma #C7527). To achieve stabilization of DD-tagged proteins in the cell lines NF54/iKD-FKBP35 and NF54/iOE-FKBP35, parasites were cultured in presence of 625 nM Shield-1 (30; 31). To induce the DiCre-mediated recombination of *loxP* sites in NF54/iKO-FKBP35 parasites, cultures were split and treated for 4 h with 100 nM rapamycin or the corresponding volume DMSO, which served as vehicle control, giving rise to FKBP35^{KO} and FKBP35^{WT} populations, respectively (32). Cultures were gassed with 3% O₂, 4% CO₂ and 93% N₂ and incubated in an airtight incubation chamber at 37°C.

4.2 Cloning of transfection constructs

CRISPR/Cas9-based gene editing of the NF54 parasites was performed using a two-plasmid approach as previously described (62; 63). This system is based on co-transfection of a plasmid that contains the expression cassettes for the Cas9 enzyme, the single guide RNA (sgRNA) and the blasticidin deaminase (BSD) resistance marker (pBF-gC), and a pD-derived donor plasmid that contains the template for the homology-directed repair of the Cas9-induced DNA double strand break (Fig. S2A) (63).

The plasmids pBF-gC_FKBP-3', pBF-gC_FKBP-5' and pBF-gC_P230p targeting the 3' or 5' end of *fkbp35* (PF3D7_1247400) or the *p230p* (PF3D7_0208900) locus (64), respectively, were generated by ligation of two annealed oligonucleotides (gRNA_{top} and gRNA_{bottom}) into the BsaI-digested pBF-gC backbone (63) using T4 DNA ligase (New England Biolabs).

The donor plasmid pD_FKBP35-iKO was generated by assembling six DNA fragments in a Gibson reaction (65) using (i) the plasmid backbone amplified by polymerase chain reaction (PCR) from pUC19 (primers PCRa_F/PCRa_R) (63), (ii/iii) the 5' and 3' homology regions amplified from genomic DNA (primers PCRb_F/PCRb_R and PCRc_F/PCRc_R, respectively), (iv) the *green fluorescent protein* (*gfp*) sequence amplified from pHcamGDV1-GFP-DD (primers PCRd_F/PCRd_R) (63), (v) the HRPII terminator amplified from genomic DNA (primers PCRe_F/PCRe_R), and (vi) the *fkbp35* sequence containing an artificial *loxPint* (33) amplified from a *P. falciparum* codon-optimized synthetic sequence (primers PCRf_F/PCRf_R) ordered from Genscript.

The donor plasmid pD_FKBP35-iKD was generated by assembling five DNA fragments in a Gibson reaction (65) using (i) the plasmid backbone amplified from pUC19 (primers PCRa_F/PCRa_R) (63), (ii/iii) the 5' and 3' homology regions amplified from genomic DNA (primers PCRg_F/PCRg_R and PCRh_F/PCRh_R, respectively), (iv) the *gfp* sequence amplified from pHcamGDV1-GFP-DD (primers PCRi_F/PCRi_R) (63), and (v) the destabilization domain (*dd*) sequence amplified from pHcamGDV1-GFP-DD (primers PCRj_F/PCRj_R) (63).

The donor plasmid pD_FKBP35-iOE was generated by assembling three DNA fragments in a Gibson reaction (65) using (i) the NheI/PstI-digested plasmid pkiwi003 (66), (ii) the *fkbp35* coding sequence amplified from genomic DNA (primers PCRk_F/PCRk_R), and (iii) the *gfp-dd* sequence amplified from pHcamGDV1-GFP-DD (primers PCRl_F/PCRl_R) (63).

Oligonucleotides are listed in Table S4.

4.3 Transfection and selection of gene-edited parasites

P. falciparum transfection using the CRISPR/Cas9 two-plasmid approach was performed as described previously (63). Briefly, 50 µg of each plasmid (pBF-gC_FKBP-3' and pD_FKBP35-iKO, pBF-gC_FKBP-5' and pD_FKBP35-iKD, pBF-gC_P230p and pD_FKBP35-iOE) were co-transfected into an NF54::DiCre line (67), in which *ap2-g* was tagged with the fluorophore mScarlet as described previously (68). Transgenic parasites were selected with 2.5 µg/mL

blastocidin-S-hydrochloride, which was added 24 h after transfection for 10 days. Transgenic populations were usually obtained 2-3 weeks after transfection and correct editing of the modified loci was confirmed by PCR on gDNA (Figs. S1B, S2B and S4B). Primer sequences used for these PCRs are listed in Table S4. Clonal parasite lines were obtained by limiting dilution cloning (69).

4.4 DNA content analysis

NF54/iKO-FKBP35 parasites were fixed in 4% formaldehyde/0.0075% glutaraldehyde for 30 min at room temperature and washed 3 times in PBS and stored at 4°C. Samples were collected from three independent biological replicates. RNA was digested by incubation with 0.1 mg/mL RNase A in PBS containing 0.1% Triton-X100 for 15 min at room temperature. Nuclei were stained using 1X SYBR Green 1 DNA stain (Invitrogen S7563) for 20 min and washed three times in PBS. The fixation step quenched the GFP signal detected by the flow cytometer (Fig. S9). SYBR Green intensity was measured using a MACS Quant Analyzer 10 and analyzed using the FlowJo_v10.6.1 software.

4.5 Western blotting

Saponin lysis of 500 µL infected red blood cells at 3-5% parasitaemia was performed by incubation for 10 min on ice in 3 mL ice-cold 0.15% saponin in PBS. The parasite pellets were washed two times in ice-cold PBS and resuspended in Laemmli buffer (62.5 mM Tris base, 2% SDS, 10% glycerol and 5% 2-mercaptoethanol). Proteins were separated on 4-12% Bis-Tris gels (Novex, Qiagen) using MOPS running buffer (Novex, Qiagen). Then proteins were transferred to a nitrocellulose membrane (GE healthcare #106000169), which was blocked with 5% milk powder in PBS/0.1% Tween (PBS-T) for 1 h. The membrane was probed using the primary antibodies mAb mouse α-GFP (1:1'000, Roche Diagnostics #11814460001), mAb mouse α-PfGAPDH (1:20'000) (70), or mAb mouse α-puromycin (1:5'000, Sigma MABE343) diluted in blocking solution. After 2 h incubation, the membrane was washed 5 times in PBS-T before it was incubated with the secondary antibody goat α-mouse IgG (H&L)-HRP (1:10'000, GE healthcare #NXA931). After 1 h incubation, the membrane was washed 5 times and the signal was detected using the chemiluminescent substrate SuperSignal West Pico Plus (Thermo Scientific, REF 34580) and the imaging system Vilber Fusion FX7 Edge 17.10 SN. Nitrocellulose membranes were stripped by incubating for 10 min in a stripping buffer (1 g/L SDS, 15 g/L glycine, 1% Tween-20, pH 2.2) and washed three times in PBS-T before blocking and re-probing.

4.6 Drug dose-response experiments

Dose-response relationships were assessed by exposing synchronous ring stage parasites at 0.5% parasitaemia and 1.25% hematocrit to 12 drug concentrations using a two-step serial dilution in 96-well plates (Corning Incorporated, 96-well cell culture plate, flat bottom, REF 3596) in three independent biological replicates. The plates were gassed and incubated for 48 h at 37 °C. FK506 (MedChemExpress HY-13756), D44 (ChemBridge 7934155 and ChemDiv 7286-2836) and cyclosporin A (Sigma 30024) stocks of 10 mM were prepared in DMSO and working solutions were prepared immediately before the experiment. To measure parasite survival using flow cytometry, cultures were stained in 1X SYBR Green 1 DNA stain (Invitrogen S7563) in a new 96-well plate (Corning Incorporated, 96-well cell culture plate, round bottom, REF 3788) and incubated in the dark for 20 min. The samples were washed once in PBS. Using a MACS Quant Analyzer 10, 50'000 events per condition were measured and analyzed using the FlowJo_v10.6.1 software. Infected RBCs were identified based on SYBR Green 1 DNA stain intensity. The gating strategy is shown in Fig. S9. To quantify parasite survival based on hypoxanthine incorporation into DNA, compounds were

dissolved in DMSO (10 mM), diluted in hypoxanthine-free culture medium and titrated in duplicates over a 64-fold range (6 step twofold dilutions) in 96 well plates. 100 μ l asexual parasite culture (prepared in hypoxanthine-free medium) were added to each well and mixed with the compound to obtain a final haematocrit of 1.25% and a final parasitemia of 0.3%. After incubation for 48 hrs, 0.25 μ Ci of [3H]-hypoxanthine was added per well and plates were incubated for an additional 24 hrs. Parasites were then harvested onto glass-fiber filters using a Microbeta FilterMate cell harvester (Perkin Elmer, Waltham, US) and radioactivity was counted using a MicroBeta2 liquid scintillation counter (Perkin Elmer, Waltham, US). The results were recorded and expressed as a percentage of the untreated controls. Curve fitting and IC50 calculations were performed using a non-linear, four-parameter regression model with variable slope (Graph Pad Prism, version 8.2.1).

4.7 Fluorescence microscopy

To localize *Pf*FKBP35, NF54/iKD-FKBP35 parasites were fixed in 4% formaldehyde/0.0075% glutaraldehyde for 30 min at room temperature and washed three times in PBS. The fixed cells were permeabilized by incubation with 0.1% Triton-X100-containing PBS for 15 min. After two washing steps in PBS, the cells were incubated in a blocking/aldehyde quenching solution (3% BSA in PBS complemented with 50 mM ammonium chloride). The samples were incubated with the primary antibodies rabbit α -GFP (1:400, Abcam ab6556) and mAb mouse α -fibrillarin (1:100, Santa Cruz Biotechnology sc-166021) for 1 h in 3% BSA in PBS and washed 3 times in PBS. Subsequently, the samples were incubated with the secondary antibodies goat α -rabbit IgG (H&L) Alexa Fluor 488 (1:250, Invitrogen A11008) and goat α -mouse IgG (H&L) Alexa Fluor 594 (1:250, Invitrogen A11032), for 1 h and washed 3 times, before they were mixed with Vectashield containing DAPI (Vector laboratories, H-1200), and mounted on a microscopy slide. Slides were imaged using a Leica THUNDER 3D Assay imaging system.

4.8 Proteomics

Parasite cultures were split at 0-6 hpi in G1 and treated for 4 h with 100 nM rapamycin. Saponin lysis of paired FKBP35^{WT} and FKBP35^{KO} populations at 36-42 hpi in G1 and 24-30 hpi in G2 was performed as described above. Samples were collected in three independent biological replicates. The washed parasite pellet was snap frozen in liquid nitrogen. After thawing, parasites were resuspended in 25 μ L of lysis buffer (5% SDS, 100 mM tetraethylammonium bromide (TEAB), 10 mM tris(2-carboxyethyl)phosphine (TCEP), pH 8.5) and sonicated (10 cycles, 30 sec on/off at 4 °C, Bioruptor, Diagenode). Lysates were subsequently reduced for 10 minutes at 95 °C. Samples were then cooled down to RT and 0.5 μ L of 1M iodoacetamide was added to the samples. Cysteine residues were alkylated for 30 min at 25 °C in the dark. Digestion and peptide purification was performed using S-trap technology (Protifi) according to the manufacturer's instructions. In brief, samples were acidified by addition of 2.5 μ L of 12% phosphoric acid (1:10) and then 165 μ L of S-trap buffer (90% methanol, 100 mM TEAB pH 7.1) was added to the samples (6:1). Samples were briefly vortexed and loaded onto S-trap micro spin-columns (Protifi) and centrifuged for 1 min at 4000 g. Flow-through was discarded and spin-columns were then washed 3 times with 150 μ L of S-trap buffer (each time samples were centrifuged for 1 min at 4000 g and flow-through was removed). S-trap columns were then moved to the clean tubes and 20 μ L of digestion buffer (50 mM TEAB pH 8.0) and trypsin (at 1:25 enzyme to protein ratio) were added to the samples. Digestion was allowed to proceed for 1h at 47 °C. Then, 40 μ L of digestion buffer were added to the samples and the peptides were collected by centrifugation at 4000 g for 1 minute. To increase the recovery, S-trap columns were washed with 40 μ L of 0.2% formic acid in water (400g, 1 min) and 35 μ L of 0.2% formic acid in 50%

acetonitrile. Eluted peptides were dried under vacuum and stored at -20 °C until further analysis.

Peptides were resuspended in 0.1% aqueous formic acid and peptide concentration was adjusted to 0.25 µg/µL. 1 µL of each sample was subjected to LC-MS/MS analysis using an Orbitrap Elicpse Tribrid Mass Spectrometer fitted with an Ultimate 3000 nano system (both from Thermo Fisher Scientific) and a custom-made column heater set to 60 °C. Peptides were resolved using a RP-HPLC column (75 µm × 30 cm) packed in-house with C18 resin (ReproSil-Pur C18-AQ, 1.9 µm resin; Dr. Maisch GmbH) at a flow rate of 0.3 µL/min. The following gradient was used for peptide separation: from 2% buffer B to 12% B over 5 min, to 30% B over 40 min, to 50% B over 15 min, to 95% B over 2 min followed by 11 min at 95% B then back to 2% B. Buffer A was 0.1% formic acid in water and buffer B was 80% acetonitrile, 0.1% formic acid in water.

The mass spectrometer was operated in DDA mode with a cycle time of 3 seconds between master scans. Each master scan was acquired in the Orbitrap at a resolution of 240,000 FWHM (at 200 m/z) and a scan range from 375 to 1600 m/z followed by MS2 scans of the most intense precursors in the linear ion trap at “Rapid” scan rate with isolation width of the quadrupole set to 1.4 m/z. Maximum ion injection time was set to 50 ms (MS1) and 35 ms (MS2) with an AGC target set to 1e6 and 1e4, respectively. Only peptides with charge state 2-5 were included in the analysis. Monoisotopic precursor selection (MIPS) was set to Peptide, and the Intensity Threshold was set to 5e3. Peptides were fragmented by HCD (Higher-energy collisional dissociation) with collision energy set to 35%, and one microscan was acquired for each spectrum. The dynamic exclusion duration was set to 30 seconds.

The acquired raw-files were imported into the Progenesis QI software (v2.0, Nonlinear Dynamics Limited), which was used to extract peptide precursor ion intensities across all samples applying the default parameters. The generated mgf-file was searched using MASCOT against a *Plasmodium falciparum* (isolate 3D7) database (Uniprot, 11.2019) and 392 commonly observed contaminants using the following search criteria: full tryptic specificity was required (cleavage after lysine or arginine residues, unless followed by proline); 3 missed cleavages were allowed; carbamidomethylation (C) was set as fixed modification; oxidation (M) and acetyl (Protein N-term) were applied as variable modifications; mass tolerance of 10 ppm (precursor) and 0.6 Da (fragments). The database search results were filtered using the ion score to set the false discovery rate (FDR) to 1% on the peptide and protein level, respectively, based on the number of reverse protein sequence hits in the dataset. Quantitative analysis results from label-free quantification were processed using the SafeQuant R package v.2.3.2 (71) to obtain peptide relative abundances. This analysis included global data normalization by equalizing the total peak/reporter areas across all LC-MS runs, data imputation using the knn algorithm, summation of peak areas per protein and LC-MS/MS run, followed by calculation of peptide abundance ratios. Only isoform specific peptide ion signals were considered for quantification. To meet additional assumptions (normality and homoscedasticity) underlying the use of linear regression models and t-Tests, MS-intensity signals were transformed from the linear to the log-scale. The summarized peptide expression values were used for statistical testing of between condition differentially abundant peptides. Here, empirical Bayes moderated t-Tests were applied, as implemented in the R/Bioconductor limma package (72). The resulting per protein and condition comparison *p*-values were adjusted for multiple testing using the Benjamini-Hochberg method.

Gene ontology enrichment analysis was performed using the PANTHER Overrepresentation Test with the correction method “False Discovery Rate”, the test type “Fisher’s Exact”, and the annotation data set “PANTHER GO-Slim” (37).

4.9 IPP complementation

Parasite cultures were split at 0-6 hpi in G1 and treated for 4 h with 100 nM rapamycin. FKBP35^{KO} and FKBP35^{WT} populations were split again and supplemented either with 200 μ M isopentenyl pyrophosphate (IPP) (Sigma I0503) or the corresponding volume H₂O. The respective culture medium was replaced daily. Parasites treated with 156 ng/mL doxycycline (Sigma D9891) were used to confirm that IPP complementation is able to rescue apicoplast-deficient parasites. DNA content analysis was performed as described above.

4.10 Heat shock susceptibility testing

Parasites cultures were split at 0-6 hpi in G1 and treated for 4 h with 100 nM rapamycin. While control populations were always incubated at 37 °C, test populations were incubated at 40 °C for 6 h starting at 24-30 hpi. Parasitaemia was measured in G2 using flow cytometry as described above.

4.11 Surface sensing of translation (SUnSET)

Parasite cultures were split at 0-6 hpi in G1 and treated for 4 h with 100 nM rapamycin. At 24-30 hpi in G2, the cultures were incubated with 1 μ g/mL puromycin (Sigma P8833) for 1 h at 37 °C (40; 73). Saponin lysis and Western blotting were performed as described above.

4.12 Cellular thermal shift assay (CETSA)

The mass spectrometry-based isothermal dose-response cellular thermal shift assay (ITDR-MS-CETSA) was performed according to the protocol developed by Dziekan and colleagues (48). Briefly, live parasites (10⁷ MACS-purified trophozoites per condition) were exposed to a FK506 (MedChemExpress HY-13756) concentration gradient (100 μ M to 1.5 nM) and a DMSO control for 1 h. Subsequently, the parasites were washed in PBS and exposed to the denaturing temperatures 50 °C, 55 °C and 60 °C as well as to a non-denaturing temperature 37 °C for 3 min, before they were cooled down to 4 °C for 3 min. Afterwards, the cells were lysed by resuspending in a lysis buffer (50 mM HEPES pH 7.5, 5 mM β -glycerophosphate, 0.1 mM Na₃VO₄, 10 mM MgCl₂, 2 mM TCEP, and EDTA-free protease inhibitor cocktail (Sigma)), three freeze/thaw cycles, and mechanical shearing using a syringe with a 31-gauge needle. Subsequently, insoluble proteins were pelleted by centrifugation (20 min at 4 °C and 20'000 g) and the soluble fraction was collected and subjected to protein digestion, reduction, alkylation, and TMT10 labelling followed by LC/MS analysis as described (48).

To prepare the protein lysate, saponin-lysed parasites were lysed in the afore-mentioned buffer by three freeze/thaw cycles and mechanical shearing. After centrifugation, soluble proteins were exposed to a FK506 concentration gradient (200 μ M to 3 nM) and a DMSO control for 1 min before the thermal challenge was performed as described above. Afterwards, the soluble protein fraction was isolated and prepared for by LC/MS analysis as described (48).

Data analysis was performed using the Proteome Discoverer 2.1 software (Thermo Fisher Scientific) and the R package “mineCETSA” (v 1.1.1). Only proteins identified by at least three peptide spectrum matches (PSMs) were included in the analysis (48).

4.13 Transcriptomics

4.13.1 Sample preparation

The RNA of 500 μ L iRBCs at 3-5% parasitaemia was isolated using 3 mL TRIzol reagent (Invitrogen) followed by purification using the Direct-zol RNA MiniPrep kit (Zymo). Samples were collected from three biological replicates. Stranded RNA sequencing libraries were prepared using the Illumina TruSeq stranded mRNA library preparation kit (REF 20020594) according to the manufacturer's protocol. The library was sequenced in 100 bp paired-end reads on an Illumina NextSeq 2000 sequencer using the Illumina NextSeq 2000 P3 reagents (REF 20040560), resulting in 5 million reads on average per sample.

4.13.2 RNA-Seq data analysis

Fastqc was run on all raw fastq files. Then, each file was aligned with hisat2 (version 2.0.5) (74) against the 3D7 reference genome (PlasmoDB v58) (75) in which the GFP and the recoded FBPK35 sequences were added at the end of chromosome 12. Read counts for each gene and for each sample were then counted using featureCount (gff file from PlasmoDB v58 modified to include the added GFP and FBPK35). TPM (Transcripts Per Kilobase Million) were then calculated from the raw count matrix.

Principal component analysis (PCA) was performed using the R (version 4.2.1) function prcomp using log transformed TPMs. PCA revealed that the rapamycin-treated replicate B of TP 3 ("3B.R") was a clear outlier and is different from any other replicate at any time point (Fig. S3D). For this reason, it was excluded from further analysis. The PCA was done again without it and the other samples show the same pattern as the initial PCA (Fig. S3E).

DESeq2 (version 1.36) (76) was used for normalization and differential expression analysis. Paired differential expression tests were done for each time point using DESeq2 LRT (likelihood ratio test) using the full model $\sim$ Treatment + Replicate (Treatment is DMSO or Rapamycin and Replicate is the clone, added as the data is paired) and the reduced model $\sim$ Replicate. From this paired test and the average log fold-change calculated from the TPMs, normalized volcano plots were made using the R package EnhancedVolcano (version 1.14) (77).

4.13.3 Cell cycle analysis

Spearman correlation between normalized scaled read counts for each sample for all genes (as one vector) and each time point of the reference microarray time course published by Bozdech *et al.* (41) (3D7 smoothed, retrieved from PlasmoDB) were tested. Spearman's coefficients rho for each time point of this RNA-Seq time course and each time point of the reference data set were plotted using ggcorrplot (version 0.1.3) (78). For each time point of this RNA-Seq dataset, the rho coefficients were then plotted against the time points of the reference data set and the FKBP35^{WT} and FKBP35^{KO} data were fitted separately to the linearized sinusoidal $A * \sin\left(\frac{2\pi}{53} \text{time}\right) + B * \cos\left(\frac{2\pi}{53} \text{time}\right) + C$ using R lm function. The time point matching best to the FKBP35^{WT} and FKBP35^{KO} samples for each time point of the RNA-Seq time course was taken as the maximum of this fitted function. Both fitted functions were F-tested (using R function var.test using both fitted model as applied to the FKBP35^{KO} data) to test whether the FKBP35^{WT} and FKBP35^{KO} data correlated with the reference data set (41) in a significantly different way, which would indicate a cell cycle shift if statistically significant. Only the last time point showed such a significant difference.

Data availability

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium repository with the dataset identifier PXD039018. The sequencing data are available in the Sequence Read Archive (SRA) via the accession number PRJNA914079.

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Author contributions

B.T.T. designed and performed experiments, analyzed and interpreted the data, and wrote the original draft of the manuscript. J.M.D. and S.T. performed and interpreted CETSA experiments. Z.B. supervised these experiments and provided resources. F.A. analyzed RNA-Seq data. M.M. supervised these experiments and provided resources for RNA-Seq data generation. A.P. performed immunofluorescence assays. K.B. performed and analyzed proteomics experiments supervised by A.S. C.Gu. and M.R. performed and interpreted hypoxanthine incorporation assays. C.Gr. edited the manuscript. N.M.B.B. conceived the study, designed and supervised experiments, provided resources, and wrote the manuscript. All authors contributed to the final editing of the manuscript.

Declaration of interests

The authors declare that they have no conflict of interest.

Supplemental figures

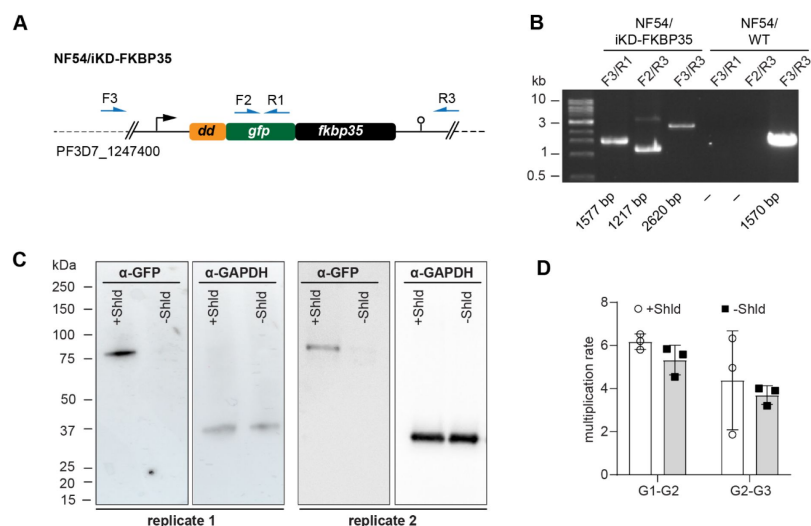


Figure S1.

Characterization of NF54/iKD-FKBP35 parasites.

(A) Schematic of the modified *fkbp35* locus (PF3D7_1247400) in NF54/iKD-FKBP35 parasites. The 5'-end of *fkbp35* was tagged with the sequences of a destabilization domain (*dd*) and a green fluorescent protein (*gfp*). The DD tag allows regulating protein levels post-translationally using the stabilizing compound Shield-1 (Shld) (30; 31). Names and binding sites of the primers used for diagnostic PCRs are indicated. **(B)** Diagnostic PCRs performed on gDNA of NF54/FKBP35 iKD parasites to confirm correct editing of the locus. PCRs per-

formed on NF54/WT gDNA served as control. Numbers at the bottom indicate expected fragment sizes. **(C)** Expression levels of the DD-GFP-FKBP35 fusion protein were assessed by Western blot. Parasite cultures were split at 0-6 hpi and treated with Shld or the vehicle control EtOH. Protein lysates were collected 40 hours later. α-GAPDH served as loading control. The expected size of the DD-GFP-FKBP35 fusion protein is 74.2 kDa. **(D)** Multiplication rates of NF54/iKD-FKBP35 parasites from generation 1 to generation 2 (G1-G2) and G2-G3. Parasites were split at 0-6 hpi and treated with Shld or the vehicle control EtOH. Multiplication rates were determined using flow cytometry. Data points represent the multiplication rates from G1-G2 and G2-G3, respectively. n=3; error bars represent the standard error of the mean.

Figure S2.

CRISPR/Cas9-based generation of transgenic parasite lines.

(A) Schematic of the wild type *fkbp35* locus (PF3D7_1247400) and the two plasmids used for CRISPR/Cas9 gene editing (pD_FKBP35-iKO and pBF-gC_FKBP-3') to generate the modified *fkbp35* locus of NF54/iKO-FKBP35 parasites. The modification consists of an intron containing a *loxP* site that was introduced after the first 38 bp of the coding sequence and an *hrpII* terminator inserted after the stop codon followed by another *loxP* site and a *gfp* sequence. The sequence flanked by the *loxP* sites was recodonized to prevent homologous recombination during gene editing. Names and binding sites of the primers used for diagnostic PCRs are indicated. **(B)** Diagnostic PCRs on gDNA of NF54/iKO-FKBP35 parasites confirm correct locus editing and rapamycin-induced *loxP* recombination. Numbers at the bottom indicate the expected band sizes. **(C)** DNA content analysis of FKBP35^{KO} and

FKBP35^{WT} parasites in presence or absence of isopentenyl pyrophosphate (IPP)-supplementation. Disruption of isoprenoid precursor biosynthesis in the apicoplast – the only essential function of this organelle – is a known cause for delayed death in *P. falciparum* (35; 36). Inhibition of isoprenoid synthesis by doxycycline can thus be restored by supplementing the parasite culture medium with the isoprenoid precursor IPP (79). However, IPP supplementation did not alter the fate of FKBP35^{KO} cells, demonstrating that their death in G2 is independent of apicoplast-dependent processes as shown by DNA content analysis. Doxycycline-treated wild-type parasites rescued by IPP-supplementation served as positive control. The knockout was induced at 0-6 hpi. Representative data of three biological replicates are shown. **(D)** Parasite survival after exposure to 40 °C for 6 h. Given the reported interactions of *Pf*FKBP35 with heat shock protein 70 (HSP70) and HSP90 (14; 16), we tested the parasites' response to heat stress. *Pf*HSP70 is required to ensure parasite survival under heat shock conditions by stabilizing the digestive vacuole membrane through binding to phosphatidylinositol 3-phosphate (80). Despite this association, we did not detect altered heat-susceptibility between FKBP35^{KO} and FKBP35^{WT} cells. The knockout was induced at 0-6 hpi. n=3, error bars represent the standard error of the mean. **(E)** Assessment of translation rates using SUnSET. FKBP35^{KO} and FKBP35^{WT} trophozoites of G1 were incubated with puromycin for 1 hour. Incorporated puromycin was detected using an α-puromycin antibody; α-GAPDH served as loading control. Replicate 3 is also shown in main figure 3 (mirrored). The control was performed using an NF54/DiCre parasite line. Untreated control parasites were cultured in absence of puromycin, confirming the specificity of the α-puromycin antibody.

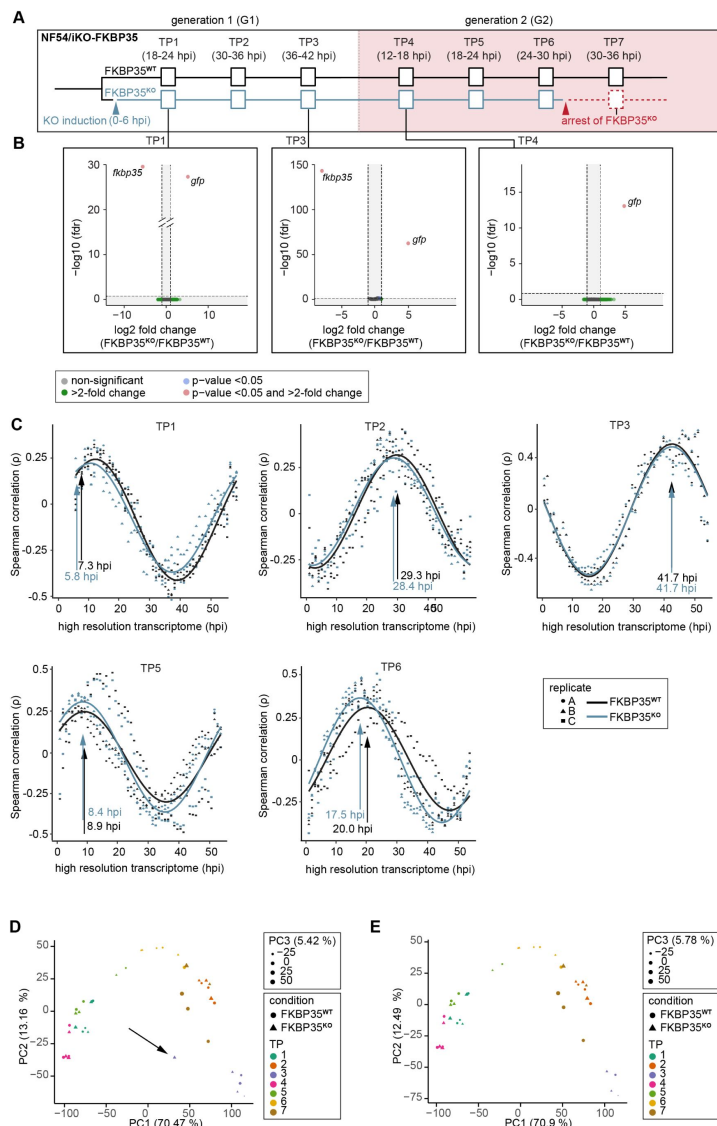


Figure S3.

Transcriptomics reveals a cell cycle arrest of FKBP35^{KO} parasites.

(A) Schematic of the experimental setup. The knock-out was induced at 0-6 hpi. RNA was collected at indicated time points (TPs). **(B)** Volcano plots show differential gene expression between FKBP35^{KO} and FKBP35^{WT} parasites. Genes are color-coded according to the fold change and *q*-value (false discovery rate), *n*=3. **(C)** Mapping of FKBP35^{KO} and FKBP35^{WT} transcriptomes to a high-resolution reference data set (41). Spearman rank coefficients (ρ) describe the correlation between the sampled transcriptomes and the reference data set. The time points with the highest correlation coefficient are indicated. **(D)** Principal component analysis (PCA) of RNA-Seq data by condition and time point (TP). The outlier at TP3 (highlighted by an arrow) was removed from the analysis. **(E)** PCA of RNA-Seq data by condition and time point without the TP3 outlier mentioned in (D). Exclusion of this sample did not affect the overall pattern.

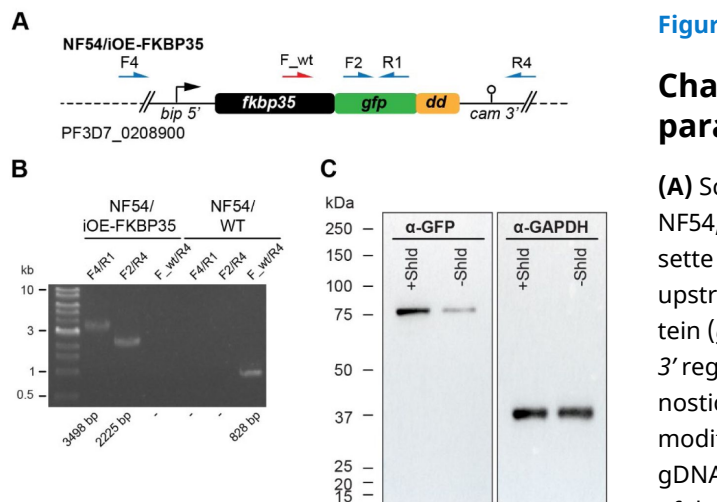


Figure S4.

Characterization of NF54/iOE-FKBP35 parasites.

(A) Schematic of the modified *p230p* locus (PF3D7_0208900) in NF54/iOE-FKBP35 parasites, in which the overexpression cassette was introduced. This cassette consists of a *bip* promoter upstream of coding sequences of *fkbp35*, green fluorescent protein (*gfp*) and destabilization domain (*dd*), followed by the *cam* 3' region. Names and binding sites of the primers used for diagnostic PCRs are indicated. The primer F_{wt} only binds the unmodified *p230p* sequence. **(B)** Diagnostic PCRs performed on gDNA of NF54/iOE-FKBP35 parasites to confirm correct editing of the locus. PCRs performed on NF54/WT gDNA served as control. Numbers at the bottom indicate the expected band sizes. **(C)** Expression levels of the FKBP35-GFP-DD fusion protein were assessed by Western blot. Parasite cultures were split at

0-6 hpi and treated with Shield or the vehicle control EtOH. Protein lysates were collected 40 h later. α -GAPDH served as loading control. The expected size of the FKBP35-GFP-DD fusion protein is 74.2 kDa.

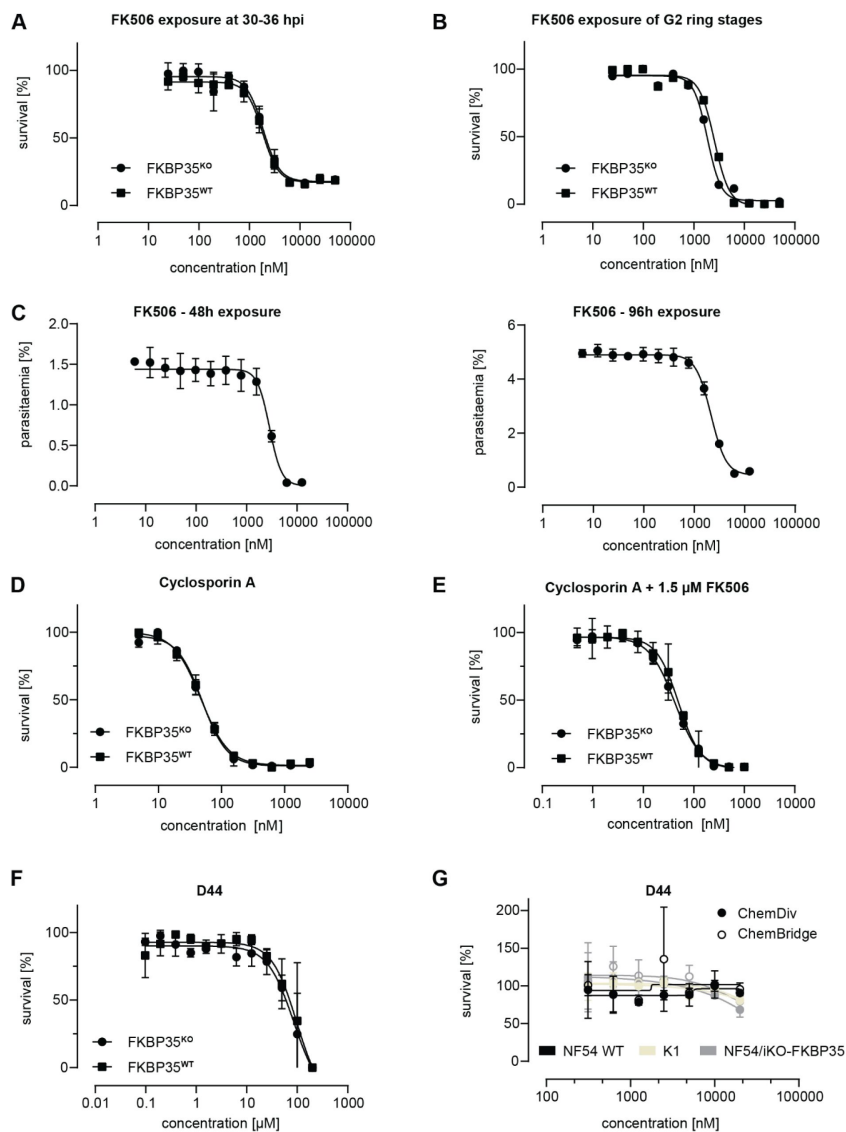


Figure S5.

FKBP-targeting drugs fail at inhibiting parasite *Pf*FKBP35.

(A) Dose-response effect of FK506 on *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT} parasites. The knockout was induced at 0-6 hpi and parasites were exposed to FK506 at 30-36 hpi. Survival rates were measured by flow cytometry in G2. $n=3$, error bars represent the standard error of the mean. **(B)** Dose-response effect of FK506 on *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT} parasites. The knockout was induced at 34-40 hpi in G1. Ring stage parasites of G2 were exposed to FK506 for 48 hours and survival rates were measured by flow cytometry in G3. Data points represent survival rates measured in a single experiment performed in technical duplicates. **(C)** Dose-response effect of FK506 on *Pf*FKBP35^{WT} parasites over two IDCs. Parasites at 0-6 hpi were exposed to FK506 and incubated for 96 hours. Parasitaemia was measured by flow cytometry after 48 hours and 96 hours. Medium containing the appropriate drug concentrations was changed after 48 hours. $n=3$, error bars represent the standard error of the mean. **(D)** Dose-response effect of cyclosporin A on *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT} parasites. Cyclosporin A, a known inhibitor of calcineurin when complexed with FKBP in

other organisms (81), did not alter drug susceptibility between *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT} parasites, suggesting that *Pf*FKBP35 is not involved in regulating calcineurin activity. The knockout was induced at 0-6 hpi. Parasites were exposed to cyclosporin A for 48 hours. $n=3$, error bars represent the standard error of the mean. **(E)** Dose-response effect of cyclosporin A in combination with a constant FK506 concentration on *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT} parasites. The knockout was induced at 0-6 hpi. Parasites were subsequently exposed to cyclosporin A and 1.5 μ M FK506 for 48 h. $n=3$, error bars represent the standard error of the mean. **(F)** Dose-response effect of D44 on *Pf*FKBP35^{KO} and *Pf*FKBP35^{WT}. The knockout was induced at 0-6 hpi and parasites were exposed to D44 for 48 hours thereafter. Survival rates were determined using flow cytometry. $n=3$, error bars represent the standard error of the mean. **(G)** Dose-response effect of D44 on NF54 WT, K1 and NF54/iKO-FKBP35 parasites after 72 h exposure. Survival rates were determined using a hypoxanthine incorporation assay. Data points represent the mean of two technical replicates. Error bars represent the standard error of the mean.

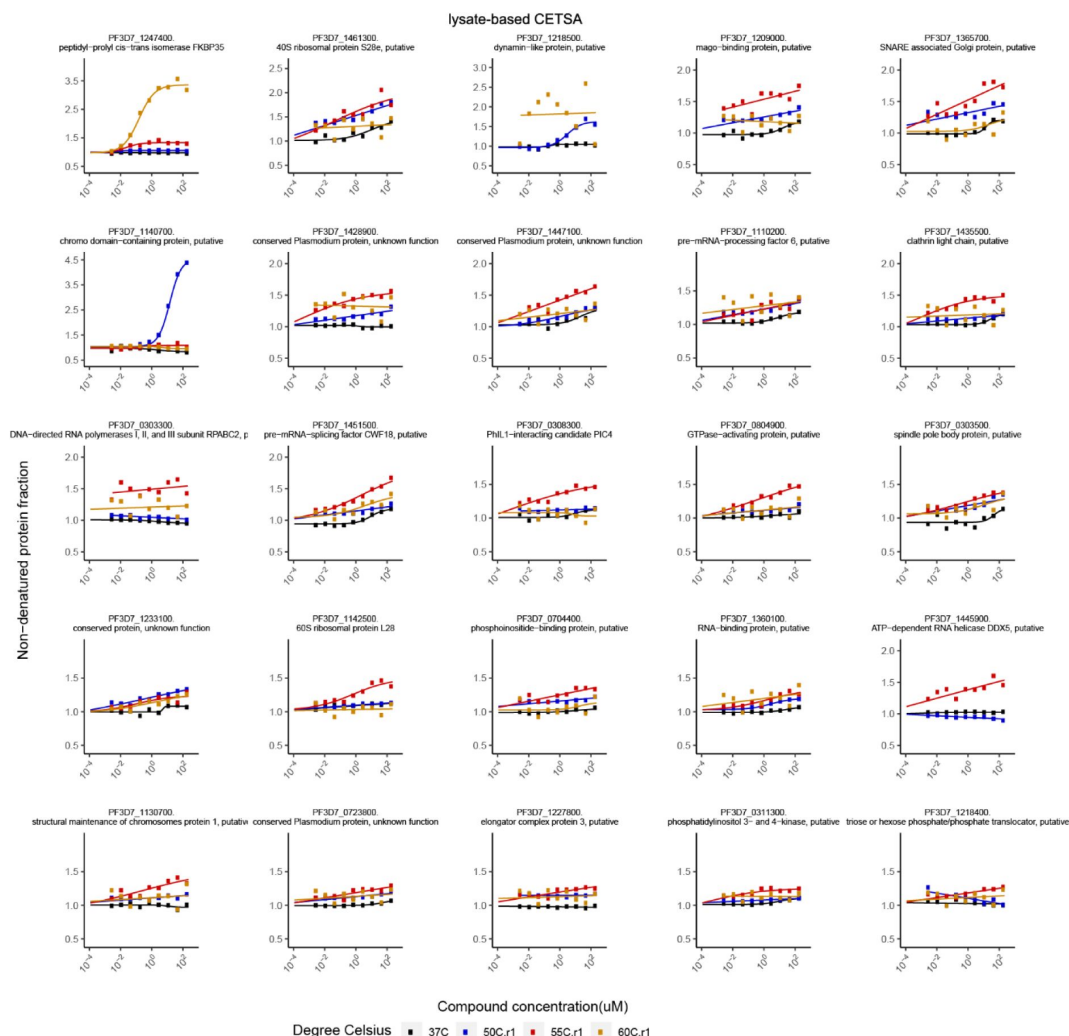


Figure S6.

Putative FK506 interaction partners identified by protein lysate-based CETSA.

Proteins with an area under the curve (ΔAUC in protein lysate-based CETSA) greater than two median absolute deviations and an R^2 greater than 0.8 are listed. Bullets represent the abundance of non-denatured proteins in response to increasing FK506 concentrations under thermal challenge relative to a DMSO control. Colors indicate the temperatures used for protein denaturation; the black curve represents the non-denaturing control.

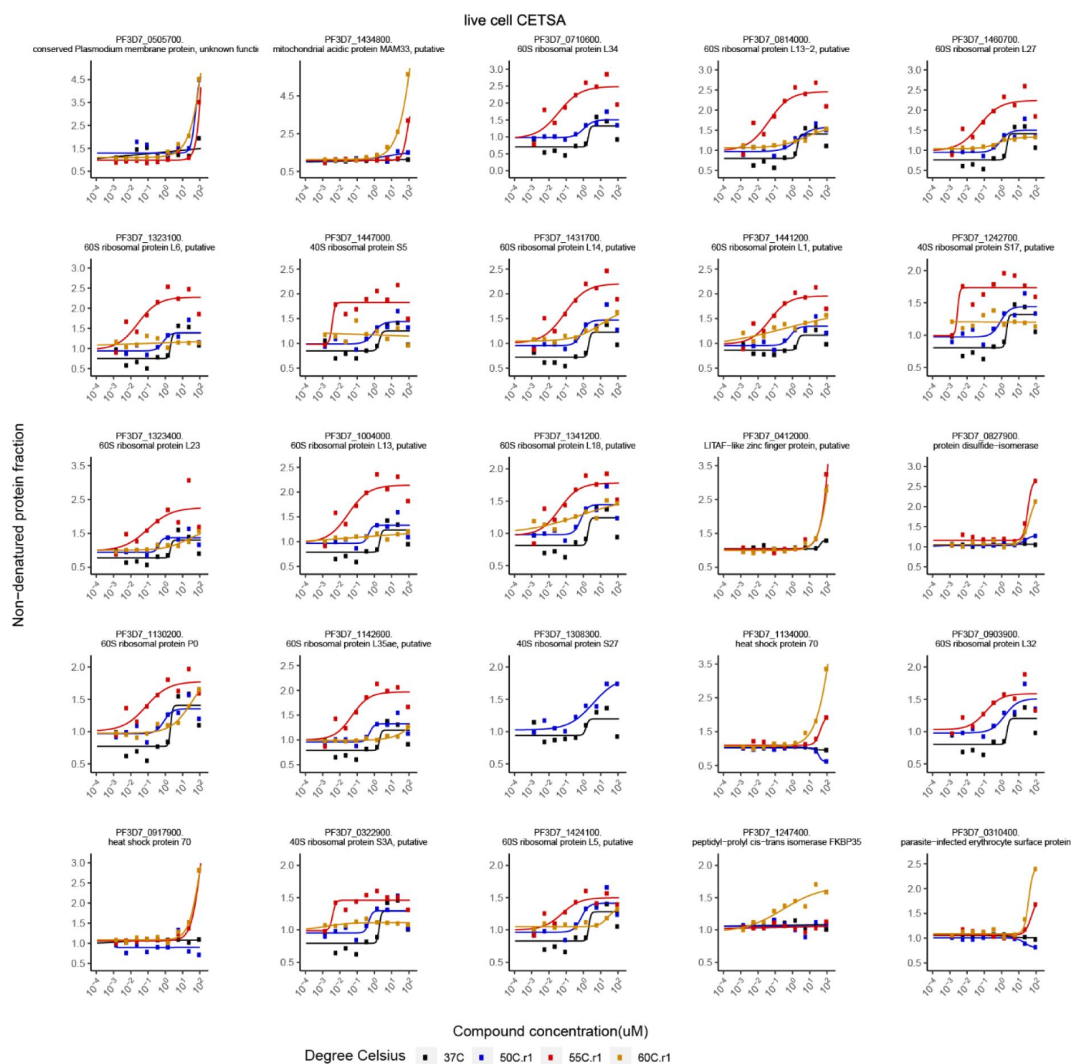


Figure S7.

Putative FK506 interaction partners identified by live cell CETSA.

Proteins with an area under the curve (Δ AUC in live cell CETSA) greater than two median absolute deviations and an R^2 greater than 0.8 are listed. Bullets represent the abundance of non-denatured proteins in response to increasing FK506 concentrations under thermal challenge relative to a DMSO control. Colors indicate the temperatures used for heat challenge; the black curve represents the non-denaturing control.

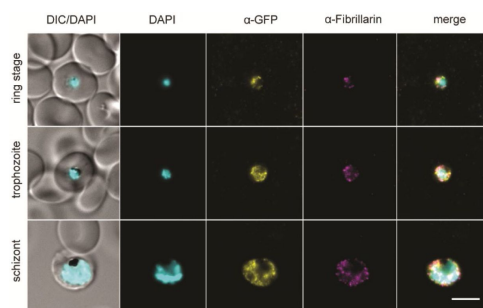


Figure S8.

***Pf*FKBP35 localizes near fibrillarin-dense nuclear regions.**

Localization of *Pf*FKBP35 (GFP) and fibrillarin in NF54/iKD-FKBP35 parasites at the ring, trophozoite and schizont stage assessed by immunofluorescence. DNA was stained with DAPI. Representative images are shown. Scale bar: 5 μ m. DIC, differential interference contrast.

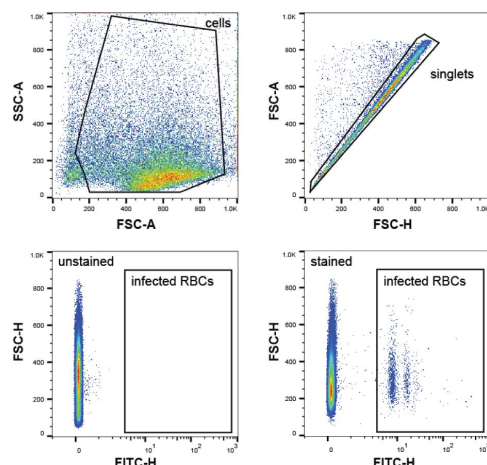


Figure S9.

Gating strategy of flow cytometry data.

Representative flow cytometry plots of formaldehyde/glutaraldehyde-fixed parasite cultures are shown. The first plot shows the gate to distinguish cells from small debris (SSC-A vs. FSC-A). The 'singlets' gate (FSC-A vs. FSC-H) removes events with more than one cells. SYBR Green fluorescence intensity is used to identify parasite-infected RBCs (FSC-H vs. FITC-H). SSC, side scatter. FSC, forward scatter. FITC, fluorescein-5-isothiocyanate. A, area. H, height.

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Editors

Reviewing Editor

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Reviewer #1 (Public Review):

In this study, the authors investigate the biological function of the FK506-binding protein FKBP35 in the malaria-causing parasite *Plasmodium falciparum*. Like its homologs in other organisms, PfkFBP35 harbors peptidyl-prolyl isomerase (PPIase) and chaperoning activities, and has been considered a promising drug target due to its high affinity to the macrolide compound FK506. However, PfkFBP35 has not been validated as a drug target using reverse genetics, and the link between PfkFBP35-interacting drugs and their antimalarial activity remains elusive. The manuscript is structured in two parts addressing the biological function of PfkFBP35 and the antimalarial activity of FK506, respectively.

The first part combines conditional genome editing, proteomics and transcriptomics analysis to investigate the effects of FKBP35 depletion in *P. falciparum*. The work is very well performed and clearly described. The data provide definitive evidence that FKBP35 is essential for *P. falciparum* blood stage growth. Conditional knockout of PfkFBP35 leads to a delayed death phenotype, associated with defects in ribosome maturation as detected by quantitative proteomics and stalling of protein synthesis in the parasite. The authors propose that FKBP35 regulates ribosome homeostasis but an alternative explanation could be that changes in the ribosome proteome are downstream consequences of the abrogation of FKBP35 essential activities as chaperone and/or PPIase. It is unclear whether FKBP35 has a specific function in *P. falciparum* as compared to other organisms. The knockdown of PfkFBP35 has no phenotypic consequence, showing that very low amounts of FKBP35 are sufficient for parasite survival and growth. In the absence of quantification of the protein during the course of the experiments, it remains unclear whether the delayed death phenotype in the knockout is due to the delayed depletion of the protein or to a delayed consequence of early protein depletion. This limitation also impacts the interpretation of the drug assays.

In the second part, the authors investigate the activity of FK506 on *P. falciparum*, and conclude that FK506 exerts its antimalarial effects independently of FKBP35. This conclusion is based on the observation that FK506 has the same activity on FKBP35 wild type and knock-out parasites, suggesting that FK506 activity is independent of FKBP35 levels, and on the fact that FK506 kills the parasite rapidly whereas inducible gene knockout results in delayed death phenotype. However, there are alternative explanations for these observations. As mentioned above, the delayed death phenotype could be due to delayed depletion of the protein upon induction of gene knockout. FK506 could have a similar activity on WT and mutant parasites when added before sufficient depletion of FKBP35 protein. In some experiments, the authors exposed KO parasites to FK506 later, presumably when the KO is effective, and obtained similar results. However, in these conditions, the death induced by the knockout could be a confounding factor when measuring the effects of the drug. Furthermore, the authors show that FK506 binds to FKBP35, and propose that the FK506-FKBP35 complex interferes with ribosome maturation, which would point towards a role of FKBP35 in FK506 action. In summary, the study does not provide sufficient evidence to rule out that FK506 exerts its effects via FKBP35.

Reviewer #2 (Public Review):

The manuscript by Thomen et al. FKBP secures ribosome homeostasis in *Plasmodium falciparum* and focuses on the importance of PfKBP35 protein, its interaction with the FK506 compound, and the role of PfKBP35 in ribosome biogenesis. The authors showed the interaction of the PfKBP54 with FK506, but the part of the FK506 and PfKBP54 in ribosome biogenesis based on the data is unclear.

The introduction is plotted with two parallel stories about PfKBP35 and FK506, with ribosome biogenesis as the central question at the end. In its current form, the manuscript suffers from two stories that are not entirely interconnected, unfinished, and somewhat confusing. Both stories need additional experiments to make the manuscript(s) more complete. The results from PfKBP35 need more evidence for the proposed ribosome biogenesis pathway control. On the other hand, the results from the drug FK506 point to different targets with lower EC50, and other follow-up experiments are needed to substantiate the authors' claims.

The strengths of the manuscript are the figures and experimental design. The combination of omics methods is informative and gives an opportunity for follow-up experiments.

Reviewer #3 (Public Review):

The study by Thommen et al. sought to identify the native role of the *Plasmodium falciparum* FKBP35 protein, which has been identified as a potential drug target due to the antiparasitic activity of the immunosuppressant FK506. This compound has multiple binding proteins in many organisms; however, only one FKBP exists in *P. falciparum* (FKBP35). Using genetically-modified parasites and mass spectrometry-based cellular thermal shift assays (CETSA), the authors suggest that this protein is involved in ribosome homeostasis and that the antiparasitic activity of FK506 is separate from its activity on the FKBP35 protein. The authors first created a conditional knockdown using the destruction domain/shield system, which demonstrated no change in asexual blood stage parasites. A conditional knockout was then generated using the DiCre system. FKBP35KO parasites survived the first generation but died in the second generation. The authors called this "a delayed death phenotype", although it was not secondary to drug treatment, so this may be a misnomer. This slow death was unrelated to apicoplast dysfunction, as demonstrated by lack of alterations in sensitivity to apicoplast inhibitors. Quantitative proteomics on the FKBP35KO vs FKBP35WT parasites demonstrated enrichment of proteins involved in pre-ribosome development and the nucleolus. Interestingly, the KO parasites were not more susceptible to cycloheximide, a translation inhibitor, in the first generation (G1), suggesting that mature ribosomes still exist at this point. The SunSET technique, which incorporates puromycin into nascent peptide chains, also showed that in G1 the FKBP35KO parasites were still able to synthesize proteins. But in the second generation (G2), there was a significant decrease in protein synthesis. Transcriptomics were also performed at multiple time points. The effects of knockout of FKBP35 were transcriptionally silent in G1, and the parasites then slowed their cell cycles as compared to the FKBP35WT parasites.

The authors next sought to evaluate whether killing by FK506 was dependent upon the inhibition of PfKBP35. Interestingly, both FKBP35KO and FKBP35WT parasites were equally susceptible to FK506. This suggested that the antiparasitic activity of FK506 was related to activity targeting essential functions in the parasite separate from binding to FKBP35. To identify these potential targets, the authors used MS-CETSA on lysates to test for thermal

stabilization of proteins after exposure to drug, which suggests drug-protein interactions. As expected, FK506 bound FKBP35 at low nM concentrations. However, given that the parasite IC₅₀ of this compound is in the uM range, the authors searched for proteins stabilized at these concentrations as putative secondary targets. Using live cell MS-CETSA, FK506 bound FKBP35 at low nM concentrations; however, in these experiments over 50 ribosomal proteins were stabilized by the drug at higher concentrations. Of note, there was also an increase in soluble ribosomal factors in the absence of denaturing conditions. The authors suggested that the drug itself led to these smaller factors disengaging from a larger ribosomal complex, leading to an increase in soluble factors. Ultimately, the authors conclude that the native function of FKBP35 is involved in ribosome homeostasis and that the antiparasitodal activity of FK506 is not related to the binding of FKBP35, but instead results from inhibition of essential functions of secondary targets.

Strengths:

This study has many strengths. It addresses an important gap in parasite biology and drug development, by addressing the native role of the potential antiparasitodal drug target FKBP35 and whether the compound FK506 works through inhibition of that putative target. The knockout data provide compelling evidence that the FKBP35 protein is essential for asexual parasite growth after one growth cycle. Analysis of the FKBP35KO line also provides evidence that the effects of FK506 are likely not solely due to inhibition of that protein, but instead must have secondary targets whose function is essential. These data are important in the field of drug development as they may guide development away from structure-based FK506 analogs that bind more specifically to the FKBP35 protein.

Weaknesses:

There are also a few notable weaknesses in the evidence that call into question the conclusion in the article title that FKBP35 is definitely involved in ribosomal homeostasis. While the proteomics supports alterations in ribosome biogenesis factors, it is unclear whether this is a direct role of the loss of the FKBP35 protein or is more related to non-specific downstream effects of knocking down the protein. The CETSA data clearly demonstrate that FK506 binds FKBP35 at low nM concentrations, which is different than the IC₅₀ noted in the parasite; however, the evidence that the proteins stabilized by uM concentrations of drug are actual targets is not completely convincing. Especially, given the high uM amounts of drug required to stabilize these proteins. This section of the manuscript would benefit from validation of a least one or two of the putative candidates noted in the text. In the live cell CETSA, it is noted that >50 ribosomal components are stabilized in drug treated but not lysate controls. Similarly, the authors suggest that the -soluble fraction of ribosomal components increases in drug-exposed parasites even at 37°C and suggests that this is likely from smaller ribosomal proteins disengaging from larger ribosomal complexes. While the evidence is convincing that this protein may play a role in ribosome homeostasis in some capacity, it is not sure that the title of the paper "FKBP secures ribosome homeostasis" holds true given the lack of mechanistic data. A minor weakness, but one that should nonetheless be addressed, is the use of the term "delayed death phenotype" with regards to the knockout parasite killing. This term is most frequently used in a very specific setting of apicomplast drugs that inhibit apicomplast ribosomes, so the term is misleading. It is also possible that the parasites are able to go through a normal cycle because of the kinetics of the knockout and that the time needed for protein clearance in the parasite to a level that is lethal.

Overall, the authors set out to identify the native role of FKBP35 in the *P. falciparum* parasites and to identify whether this is, in fact, the target of FK506. The data clearly demonstrate that FKBP35 is essential for parasite growth and provide evidence that alterations in its levels have proteomic but not transcriptional changes. However, the conclusion that FKBP35

actually stabilizes ribosomal complexes remains intermediate. The data are also very compelling that FK506 has secondary targets in the parasite aside from FKBP35; however, the high μM concentrations of the drug needed to attain results and the lack of biological validation of the CETSA hits makes it difficult to know whether any of these are actually the target of the compound or instead are nonspecific downstream consequences of treatment.

Author Response

***eLife* assessment**

This important study addresses both the native role of the Plasmodium falciparum protein PfFKBP35 and whether this protein is the target of FK506, an immunosuppressant with antiparasitic activity. The genetic evidence for the essentiality of FKBP35 in parasite growth is compelling. However, the conclusion that the role of FKBP35 is to secure ribosome homeostasis and the claim that FK506 exerts its antimalarial activity independently of FKBP35 rely on incomplete evidence.

We thank the Reviewers and Editors for their careful evaluation of our manuscript and the constructive criticism. We realized that some of our conclusions may be regarded/misunderstood as overstatements. This was by no means our intention and we apologize for the unnecessary inconvenience. The phenotype of FKBP35 knock-out parasites clearly centers on failing ribosomes and protein synthesis, which in our opinion, provides an important leap towards understanding the role of this drug target in *P. falciparum* biology. It is however correct that, at this point, we can only make evidence-based hypotheses about direct interaction partners and we will emphasize this more clearly in a revised version of the manuscript. In order to prevent misinterpretation of our work, and as detailed in the point-by-point responses to the reviewer comments, we propose changing the manuscript title to “Genetic validation of _Pf_FKBP35 as an antimalarial drug target”. To address the criticism regarding the effects of FK506, we will perform specific additional experiments. We are convinced that this new data set will resolve any remaining ambiguities and allows for a conclusive assessment of FK506 drug activity in *P. falciparum*.

Reviewer #1 (Public Review):

In this study, the authors investigate the biological function of the FK506-binding protein FKBP35 in the malaria-causing parasite Plasmodium falciparum. Like its homologs in other organisms, PfFKBP35 harbors peptidyl-prolyl isomerase (PPIase) and chaperoning activities, and has been considered a promising drug target due to its high affinity to the macrolide compound FK506. However, PfFKBP35 has not been validated as a drug target using reverse genetics, and the link between PfFKBP35-interacting drugs and their antimalarial activity remains elusive. The manuscript is structured in two parts addressing the biological function of PfFKBP35 and the antimalarial activity of FK506, respectively.

*The first part combines conditional genome editing, proteomics and transcriptomics analysis to investigate the effects of FKBP35 depletion in *P. falciparum*. The work is very well performed and clearly described. The data provide definitive evidence that FKBP35 is essential for *P. falciparum* blood stage growth. Conditional knockout of PfFKBP35 leads to a delayed death phenotype, associated with defects in ribosome maturation as detected by quantitative proteomics and stalling of protein synthesis in the parasite. The authors propose that FKBP35 regulates ribosome homeostasis but an alternative explanation could be that changes in the ribosome proteome are downstream consequences of the abrogation of FKBP35 essential activities as chaperone and/or PPIase. It is unclear whether FKBP35 has a specific function in *P. falciparum* as compared to other organisms. The knockdown of PfFKBP35 has no phenotypic consequence, showing that very*

low amounts of FKBP35 are sufficient for parasite survival and growth. In the absence of quantification of the protein during the course of the experiments, it remains unclear whether the delayed death phenotype in the knockout is due to the delayed depletion of the protein or to a delayed consequence of early protein depletion. This limitation also impacts the interpretation of the drug assays.

We thank the Reviewer for the compliments regarding our experimental setup and the clarity of our manuscript. We agree that the link between FKBP35 knock-out and ribosome homeostasis is indirect and we now emphasize this more clearly in the revised manuscript. To prevent a general misinterpretation of our manuscript, we will adapt the title accordingly.

We would still like to reiterate that the phenotype of FKBP35 knock-out parasites is best described by their defects in maintaining functional ribosomes. It is for several reasons that we believe the links between FKBP35 and ribosome function are purely evidence driven: First, pre-ribosomal and nucleolar factors are the first proteins (in generation 1 schizonts) to be affected upon knock-out of *fkbp35* (Figure 2A, Table S1). We realized that Figure 2A falls short in showing this observation, which is why will update the figure accordingly. Second, the dysregulation of ribosomal factors and the general stall in protein synthesis is dominating the phenotype of FKBP35 knock-out parasites in generation 2. We thus believe it is appropriate to say that knock-out cells are most likely killed in response to defective ribosome maintenance – which is a consequence of reduced FKBP35 levels. We are aware that our experiments (and possibly any other reverse genetics approach) cannot rule out that FKBP35 affects ribosomal factors indirectly. Clearly, more work is required to disentangle this question in more detail in the future.

We agree with the Reviewer that it is not possible to tell if the delayed death-like phenotype is due to a “delayed protein depletion”. We would however like to note that the DiCre/loxP approach allows for an immediate knock-out at the genome level and is thus as precise as possible. Further, in addition to the substantial depletion of FKBP35 in knock-out cells during the phenotypically silent generation, knocking out of *fkbp35* at earlier time points (TPs 24-30 and 34-40 hpi in the preceding generation) resulted in the very same phenotype cycle (Figure 1). Here, parasite death was delayed substantially longer, i.e. more than one complete cycle. Together with the dysregulation of early ribosome maturation in generation 1, these findings point towards a delayed death phenotype. It is of course still possible to explain the delayed death-like phenotype by remnant activity of proteins synthesized prior to the genomic knock-out. We address this possibility and describe the two scenarios mentioned by the Reviewer in lines 141-144. Disentangling the two possibilities in future experiments will be difficult, not only with regards to FKBP35, but regarding “delayed death” phenotypes in general.

*In the second part, the authors investigate the activity of FK506 on *P. falciparum*, and conclude that FK506 exerts its antimalarial effects independently of FKBP35. This conclusion is based on the observation that FK506 has the same activity on FKBP35 wild type and knock-out parasites, suggesting that FK506 activity is independent of FKBP35 levels, and on the fact that FK506 kills the parasite rapidly whereas inducible gene knockout results in delayed death phenotype. However, there are alternative explanations for these observations. As mentioned above, the delayed death phenotype could be due to delayed depletion of the protein upon induction of gene knockout. FK506 could have a similar activity on WT and mutant parasites when added before sufficient depletion of FKBP35 protein. In some experiments, the authors exposed KO parasites to FK506 later, presumably when the KO is effective, and obtained similar results. However, in these conditions, the death induced by the knockout could be a confounding factor when measuring the effects of the drug. Furthermore, the authors show that FK506 binds to FKBP35, and propose that the FK506-FKBP35 complex interferes with ribosome maturation, which would point towards a role of FKBP35 in FK506 action. In summary, the study does not provide sufficient evidence to rule out that FK506 exerts its effects via FKBP35.*

Noteworthy, we were also very much surprised by data indicating that the antimalarial activity of FK506 is independent of FKBP35. It is for this reason that we conducted a comprehensive set of experiments to disprove our initial observations, but couldn't find any evidence for an FKBP35-dependent mode of action of FK506:

We were not able to see altered FK506 sensitivity in (i) inducible knock-down parasites, (ii) inducible overexpression parasites and (iii) inducible knock-out parasites. Parasites with altered FKBP35 levels (as assessed by Western blot and quantitative proteomics at 36-42 hpi, respectively) were equally sensitive to FK506. Importantly, at no sub-lethal FK506 concentration did lower FKBP35 levels lead to an altered response of FKBP35KO compared to the wild-type control population. Furthermore, (iv) induction of the knock-out in the cycle preceding FK506 exposure also had no effect on parasite sensitivity. As mentioned by the Reviewer, we also exposed the parasites to FK506 at 30-36 hpi and (v) did not see any effect, even though we measured a 19-fold difference in FKBP35 protein levels between the parasite populations at 36-42 hpi. At this point, parasite death induced by the knock-out cannot be a confounding factor (as it was mentioned by the Reviewer), because the FKBP35 knock-out has no effect on parasite survival in generation 1 in the absence of FK506 (Figure 1F). This demonstrates that the observed effect is only due to drug-mediated killing and not due to the FKBP35 knock-out.

To account for a scenario in which the drop in FKBP35 levels only occurs after 36 hpi, we will perform an additional set of experiments, in which we induce the knock-out at 0-6 hpi and treat the parasites at 36-42 hpi (i.e. the time point at which the 19-fold difference in protein levels was measured by quantitative proteomics). This setup will allow determining whether or not the parasite killing activity of FK506 depends on FKBP35 levels.

So far, our experiments cannot support any scenario in which FK506 kills *P. falciparum* parasites via inhibiting the essential role of FKBP35 and we would therefore want to insist that this statement is based on highly solid evidence. In this context, it is important to note that our conclusion includes two scenarios: "This indicates that either the binding of FK506 does not interfere with the essential role of _Pf_FKBP35, or that _Pf_FKBP35 is inhibited only at high FK506 concentrations that also inhibit other essential factors." While this phrase is already present in our initial submission, we will emphasize this point more clearly in the revised manuscript. We are convinced that this information is of high importance for ongoing and future drug development.

Reviewer #2 (Public Review):

The manuscript by Thomen et al. FKBP secures ribosome homeostasis in Plasmodium falciparum and focuses on the importance of PfKBP35 protein, its interaction with the FK506 compound, and the role of PfKBP35 in ribosome biogenesis. The authors showed the interaction of the PfKBP54 with FK506, but the part of the FK506 and PfKBP54 in ribosome biogenesis based on the data is unclear.

The introduction is plotted with two parallel stories about PfKBP35 and FK506, with ribosome biogenesis as the central question at the end. In its current form, the manuscript suffers from two stories that are not entirely interconnected, unfinished, and somewhat confusing. Both stories need additional experiments to make the manuscript(s) more complete. The results from PfKBP35 need more evidence for the proposed ribosome biogenesis pathway control. On the other hand, the results from the drug FK506 point to different targets with lower EC50, and other follow-up experiments are needed to substantiate the authors' claims.

The strengths of the manuscript are the figures and experimental design. The combination of omics methods is informative and gives an opportunity for follow-up experiments.

We thank the Reviewer for the evaluation of the manuscript. We apologize for the fact that the Reviewer found the manuscript to be inaccessible. We will use the comments as an incentive to restructure the manuscript and do our best to clarify the presentation, interpretation and conclusion of the presented data in the revised version. We believe that the FKBP35 data are strongly interlinked with the findings on FK506. We will emphasize these links more clearly and are convinced that the complementary nature of the datasets are a particular strength of the presented work.

Reviewer #3 (Public Review):

The study by Thommen et al. sought to identify the native role of the Plasmodium falciparum FKBP35 protein, which has been identified as a potential drug target due to the antiparasitic activity of the immunosuppressant FK506. This compound has multiple binding proteins in many organisms; however, only one FKBP exists in P. falciparum (FKBP35). Using genetically-modified parasites and mass spectrometry-based cellular thermal shift assays (CETSA), the authors suggest that this protein is involved in ribosome homeostasis and that the antiparasitic activity of FK506 is separate from its activity on the FKBP35 protein. The authors first created a conditional knockdown using the destruction domain/shield system, which demonstrated no change in asexual blood stage parasites. A conditional knockout was then generated using the DiCre system. FKBP35KO parasites survived the first generation but died in the second generation. The authors called this "a delayed death phenotype", although it was not secondary to drug treatment, so this may be a misnomer. This slow death was unrelated to apicoplast dysfunction, as demonstrated by lack of alterations in sensitivity to apicoplast inhibitors. Quantitative proteomics on the FKBP35KO vs FKBP35WT parasites demonstrated enrichment of proteins involved in pre-ribosome development and the nucleolus. Interestingly, the KO parasites were not more susceptible to cycloheximide, a translation inhibitor, in the first generation (G1), suggesting that mature ribosomes still exist at this point. The SunSET technique, which incorporates puromycin into nascent peptide chains, also showed that in G1 the FKBP35KO parasites were still able to synthesize proteins. But in the second generation (G2), there was a significant decrease in protein synthesis. Transcriptomics were also performed at multiple time points. The effects of knockout of FKBP35 were transcriptionally silent in G1, and the parasites then slowed their cell cycles as compared to the FKBP35WT parasites.

The authors next sought to evaluate whether killing by FK506 was dependent upon the inhibition of PfFKBP35. Interestingly, both FKBP35KO and FKBP35WT parasites were equally susceptible to FK506. This suggested that the antiparasitic activity of FK506 was related to activity targeting essential functions in the parasite separate from binding to FKBP35. To identify these potential targets, the authors used MS-CETSA on lysates to test for thermal stabilization of proteins after exposure to drug, which suggests drug-protein interactions. As expected, FK506 bound FKBP35 at low nM concentrations. However, given that the parasite IC₅₀ of this compound is in the μM range, the authors searched for proteins stabilized at these concentrations as putative secondary targets. Using live cell MS-CETSA, FK506 bound FKBP35 at low nM concentrations; however, in these experiments over 50 ribosomal proteins were stabilized by the drug at higher concentrations. Of note, there was also an increase in soluble ribosomal factors in the absence of denaturing conditions. The authors suggested that the drug itself led to these smaller factors disengaging from a larger ribosomal complex, leading to an increase in soluble factors. Ultimately, the authors conclude that the native function of FKBP35 is involved in ribosome homeostasis and that the antiparasitic activity of FK506 is not related to the binding of FKBP35, but instead results from inhibition of essential functions of secondary targets.

Strengths:

This study has many strengths. It addresses an important gap in parasite biology and drug development, by addressing the native role of the potential antiparasitic drug target FKBP35

and whether the compound FK506 works through inhibition of that putative target. The knockout data provide compelling evidence that the FKBP35 protein is essential for asexual parasite growth after one growth cycle. Analysis of the FKBP35KO line also provides evidence that the effects of FK506 are likely not solely due to inhibition of that protein, but instead must have secondary targets whose function is essential. These data are important in the field of drug development as they may guide development away from structure-based FK506 analogs that bind more specifically to the FKBP35 protein.

Weaknesses:

There are also a few notable weaknesses in the evidence that call into question the conclusion in the article title that FKBP35 is definitely involved in ribosomal homeostasis. While the proteomics supports alterations in ribosome biogenesis factors, it is unclear whether this is a direct role of the loss of the FKBP35 protein or is more related to non-specific downstream effects of knocking down the protein. The CETSA data clearly demonstrate that FK506 binds PfFKB35 at low nM concentrations, which is different than the IC₅₀ noted in the parasite; however, the evidence that the proteins stabilized by μ M concentrations of drug are actual targets is not completely convincing. Especially, given the high μ M amounts of drug required to stabilize these proteins. This section of the manuscript would benefit from validation of a least one or two of the putative candidates noted in the text. In the live cell CETSA, it is noted that >50 ribosomal components are stabilized in drug treated but not lysate controls. Similarly, the authors suggest that the -soluble fraction of ribosomal components increases in drug-exposed parasites even at 37°C and suggests that this is likely from smaller ribosomal proteins disengaging from larger ribosomal complexes. While the evidence is convincing that this protein may play a role in ribosome homeostasis in some capacity, it is not sure that the title of the paper "FKBP secures ribosome homeostasis" holds true given the lack of mechanistic data. A minor weakness, but one that should nonetheless be addressed, is the use of the term "delayed death phenotype" with regards to the knockout parasite killing. This term is most frequently used in a very specific setting of apicoplast drugs that inhibit apicoplast ribosomes, so the term is misleading. It is also possible that the parasites are able to go through a normal cycle because of the kinetics of the knockout and that the time needed for protein clearance in the parasite to a level that is lethal.

Overall, the authors set out to identify the native role of FKBP35 in the *P. falciparum* parasites and to identify whether this is, in fact, the target of FK506. The data clearly demonstrate that FKBP35 is essential for parasite growth and provide evidence that alterations in its levels have proteomic but not transcriptional changes. However, the conclusion that FKBP35 actually stabilizes ribosomal complexes remains intermediate. The data are also very compelling that FK506 has secondary targets in the parasite aside from FKBP35; however, the high μ M concentrations of the drug needed to attain results and the lack of biological validation of the CETSA hits makes it difficult to know whether any of these are actually the target of the compound or instead are nonspecific downstream consequences of treatment.

We appreciate the detailed and valuable suggestions to improve the manuscript. We agree that CETSA could only identify potential targets of FK506 in the micromolar range, while FK506 showed a high affinity for FKBP35, consistent with earlier reports (2). We would however like to point out that FK506 kills *P. falciparum* at exactly these relatively high concentrations and not at those presumed from the high affinity interactions between FK506 and FKBP35. The relatively high FK506 concentration required to stabilize potential off target proteins is therefore not a concerning observation, but rather corroborates our conclusion that FK506 fails to inhibit the essential function of FKBP35 at concentrations that leave off targets unaffected. As mentioned in response to Reviewer 1, we will describe and discuss these data more clearly in the revised manuscript.

We thank the Reviewer for pointing out the potential issues regarding the use of the term "delayed death phenotype". We now refer to the FKBP35 phenotype as "delayed death-like" in

the revised manuscript.

We believe that follow-up work on specific FK506 CETSA hits is out of scope of the current and already quite complex manuscript.

As mentioned in the response to Reviewer 1, we realize that the short title of the manuscript can be regarded as an overstatement. Again, this was clearly not our intention and we apologize that the Reviewers had to indicate this issue. While we believe that the message of the title holds true (see response to Reviewer 1), we recognize the misconception that might arise from it, which is why we propose the new title: “Genetic validation of _Pf_FKBP35 as an antimalarial drug target”.

1. Kennedy K, Cobbold SA, Hanssen E, Birnbaum J, Spillman NJ, McHugh E, et al. Delayed death in the malaria parasite *Plasmodium falciparum* is caused by disruption of prenylation-dependent intracellular trafficking. *PLoS Biol.* 2019;17(7):e3000376.
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3. Kasahara K, Nakayama R, Shiwa Y, Kanesaki Y, Ishige T, Yoshikawa H, et al. Fpr1, a primary target of rapamycin, functions as a transcription factor for ribosomal protein genes cooperatively with Hmo1 in *Saccharomyces cerevisiae*. *PLoS Genet.* 2020;16(6):e1008865.